MULTIDRUG RESISTANT GRAM-POSITIVE BACTERIA IN TIGERNUT MILK

BY

CLEMENT ISAIAH OYINLO
210807523

A PROJECT SUBMITTED IN PARTIAL FULFILMENT OF THE REQUIREMENT FOR AWARD OF BACHELOR OF SCIENCE (B.Sc.) DEGREE IN MICROBIOLOGY, FACULTY OF SCIENCE, UNIVERSITY OF LAGOS, AKOKA- YABA, LAGOS.

SEPTEMBER, 2025

MULTIDRUG RESISTANT GRAM-POSITIVE BACTERIA IN TIGERNUT MILK

BY

CLEMENT ISAIAH OYINLO
210807523

A PROJECT SUBMITTED IN PARTIAL FULFILMENT OF THE REQUIREMENT FOR AWARD OF BACHELOR OF SCIENCE (B.Sc.) DEGREE IN MICROBIOLOGY, FACULTY OF SCIENCE, UNIVERSITY OF LAGOS, AKOKA- YABA, LAGOS.

SEPTEMBER, 2025

PROF. R.E. UZEH	**PROF. C.I. AYOLABI**
SUPERVISOR	**HEAD OF DEPARTMENT**
DATE	**DATE**

CERTIFICATION

This is to certify that this project work: **Multidrug-Resistant Gram-positive Bacteria in Tigernut Milk** was duly carried out by **CLEMENT ISAIAH OYINLO** with Matriculation Number **210807523** of the Department Of Microbiology under my supervision.

_____ _____

PROF. R.E. UZEH **DATE**

SUPERVISOR

DEDICATION

I dedicate this work to my parents Mr. and Mrs. Clement for their financial, moral and spiritual support. I also dedicate it to my brother and my sponsor Mr. Clement Abraham for believing even when I doubted myself, and also to God.

ACKNOWLEDGMENT

First and foremost, all glory be to Almighty God, who has guided me through my academic journey and granted me the strength to complete this project.

I wish to express my sincere appreciation to the Head of Department, Professor C. I. Ayolabi, my supervisor, Prof. R. E. Uzeh, and all the lecturers as well as the non-academic staff of the Department of Microbiology, University of Lagos, for the knowledge, guidance, and support they have imparted in me throughout my studies. I am especially grateful to my supervisor, Prof. R. E. Uzeh, for her invaluable guidance, patience, encouragement, and constructive criticism throughout the course of this research.My heartfelt thanks also go to Mr. Nwabueze C. W., one of the laboratory technologists in the Microbiology Laboratory, Faculty of Science, for his technical assistance and support during the experimental work.

I am profoundly grateful to my wonderful and supportive parents, Mr. and Mrs. Clement, for their financial, moral, and emotional support in every way. I also deeply appreciate my brother and sponsor, Clement Abraham, for his care and generosity, as well as my amazing siblings for their love, encouragement, and patience.Special thanks to my cooperative project members—Haleemah, Olamide, Chisom, Odunleye, Ife, and Fortune—for making this work easier and more enjoyable than I anticipated. I would also like to extend my appreciation to my dear friends—Fadilat, Farouk, Fuad, Anthony, Daniel, Wisdom, Ire and Ayo—for their constant support and encouragement.

To everyone who contributed to the success of this work in one way or another, I say a very big thank you.

TABLE OF CONTENTS

Title Page

Certification iii

Dedication iv

Acknowledgment v

Table of Contents vi

Appendices ix

List of Figures x

List of Tables xii

Abstract xiii

CHAPTER ONE 1

1.0 Introduction and Literature Review 1

1.1 Background of Study 1

1.2 Literature Review 4

1.2.1 Origin and distribution Of tigernut 4

1.2.2 Classification of tigernut 5

1.2.3 Tigernut Composition 5

1.2.4 Products obtained from tigernut 6

1.2.5 Preparation of tigernut milk (locally) 6

1.2.6 Sources of microorganism in tigernut milk 8

1.2.7 Gram-positive bacteria isolated from tigernut milk 8

1.2.8 *Bacillus cereus* 9

1.2.9 *Staphylococcus aureus* 9

1.3 Enterococci 10

1.3.1 Factors Contributing to Microbial growth in food 11

1.3.2 Antibiotic Resistance in Foodborne Gram-Positive Bacteria 13

1.3.3 Public Health Implications 13

1.4 Antibiotic Resistant Gram-positive Bacteria in Tigernut Milk 13

1.5 Statement of Problem 15

1.6 Significance of Study 15

1.7 Aim of Study 15

1.8 Objectives of Study 16

CHAPTER TWO 17

2.1 Materials and Methods 17

2.1.1 Equipment and Apparatus used 17

2.1.2 Reagents and Media used 17

2.2 Sterilization of materials 17

2.3 Sample collection 17

2.4 Microbiological analysis 18

2.4.1 Media preparation 18

2.4.2 Serial dilution 18

2.4.3 Plating methods 18

2.5 Characterization and identification of bacterial isolates 18

2.5.1 Colonial Morphology 18

2.6 Identification and Biochemical test of bacterial Isolates 18

2.6.1 Gram's stain 18

2.6.2 Catalase test 19

2.6.3 Coagulase test 19

2.6.4 Motility test 19

2.6.5 Citrate test 19

2.6.5 Salt tolerance test 19

2.7 Antimicrobial susceptibility test 20

CHAPTER THREE 21

RESULTS 21

CHAPTER FOUR 32

DISCUSSION 32

4.1 Conclusion And Recommendation 34

REFERENCES 35

APPENDIX I 39

APPENDIX II 43

APPENDIX III 50

APPENDIX IV 62

APPENDIX V 63

APPENDIX VI 64

APPENDIX VII 67

APPENDICES

		Page
Appendix I:	Total bacterial count of samples on selective media	39
Appendix II:	Statistical analysis of bacterial load on branded and unbranded tigernut milk samples on selective media	43
Appendix III:	Statistical analysis on antibiotics susceptibility test of bacterial isolates	50
Appendix IV:	Occurrence of Gram-positive bacterial in tigernut milk sample	62
Appendix V:	Multidrug resistance pattern of Gram-positive bacterial isolates from tigernut milk	63
Appendix VI:	Pictures from methodology	64
Appendix VII:	Composition of media	67

LIST OF FIGURES

		Page
Figure 1.1:	Tigernut	4
Figure 1.2:	Flow chart for traditional preparation of Tigernut milk	7
Figure 3.1:	Comparison of bacterial load in branded and unbranded tigernut milk on Manittol Salt Agar	22
Figure 3.2:	Comparison of bacterial load in branded and unbranded tigernut milk on Bile Esculin Agar	23
Figure 3.3:	Comparison of bacterial load in branded and unbranded tigernut milk on *Bacillus cereaus* agar	24
Figure 3.4:	Occurrence of Gram-positive bacteria in tigernut milk	27
Figure 3.5:	Antibiotic resistance pattern of bacterial isolates	30
Figure 3.6:	Multidrug resistance pattern of Gram-positive bacteria isolates from tigernut milk	31

LIST OF TABLES

		Page
Table 3.1:	Cultural characteristics of bacterial colonies from tigernut milk samples	25
Table 3.2:	Morphiological and biochemical characteristics of bacterial isolates from tigernut milk samples	26
Table 3.3:	Susceptibility profile for Gram-positive multidrug resistant bacteria in tigernut milk	28

ABSTRACT

Tigernut milk is a widely consumed non-alcoholic, plant-based beverage valued for its nutritional and health benefits. However, like many ready-to-drink products, it is susceptible to microbial contamination, which poses potential public health risks. This study aims to determine the prevalence and multidrug resistance patterns of Gram-positive bacteria in tigernut milk purchased from different locations in Lagos, Nigeria.Twenty-Five (25) branded, and unbranded tigernut milk samples were purchased from various vendors across selected locations in Lagos. The samples were cultured on selective media to isolate Gram-positive bacteria, which were then identified using standard microbiological and biochemical methods. Confirmed isolates were subjected to antibiotic susceptibility testing using the Kirby-Bauer disk diffusion method, antibiotics used include; amoxicllin, erythromycin, azithromycin, ofloxacin, ciprofloxacin, perfloxacin, cotrimazole, ceftriazone, levofloxacin and tetracycline. A total of 20 Gram-positive isolates were identified, predominantly as *Staphylococcus aureus*, *Enterococcus faecalis*, and *Bacillus cereus*. Multidrug resistance was observed in 85% of isolates. The prevalence of resistance, particularly to antibiotics such as erythromycin (80%), amoxicillin(75%), cotrimazole (70%) and ceftriazone (95%), highlights the multidrug resistance of the isolates .Tigernut milk serves as a potential vehicle for the transmission of multidrug-resistant Gram-positive bacteria, posing a serious public health concern. The result obtained showed high bacterial load counts which ranged from 1.225×10^8 to 2.2×10^8 cfu/ml .Among the 20 isolates , *Staphylococcus aureus* and *Enterococcus faecalis* were most prevalent with 35% of isolates followed by *Bacilllus cereus* 30%.This investigation elucidated that tigernut milk is tainted with multidrug-resistant Gram-positive bacteria, prominently including *S. aureus*, *E. faecalis*, and *B. cereus*. The elevated rates of resistance observed against widely utilized antibiotics present a significant public health hazard. Consequently, the enhancement of hygiene practices during production, more rigorous oversight of street-vended beverages, and ongoing surveillance of antimicrobial resistance are imperative to protect consumers

CHAPTER ONE

1.0 INTRODUCTION AND LITERATURE REVIEW

1.1 BACKGROUND OF STUDY

Kunu is a fermented non-alcoholic beverage popular in the northern part of Nigeria and consumed mostly by non-alcoholic drink lovers due to it refreshing organoleptic property . Kunu generally,is a non-alcoholic beverage made from millet and sorghum in Nigeria . Kunu is of various types depending on the major cereal and other materials used in its production and this include; kunu-zaki, kunu -gyada, kunu-tsamiya, kunu-akamu, kunu-aya, kunu-koko, kunu-aduwa, kunu-acha, kunu-kanwa, kunu-jiko, kunu-banle, kunu-gayamba and kunu-amshau (Ndukwe *et al*., 2023).

Milk is the whitish secretion of mammary glands of healthy mammals such as goat, cow , sheep, etc. It is acidic with pH range of 6.5 - 6.7 . It contain diverse source of nutrients; water, carbohydrate, mineral, protein, vitamin, which is essential for maintaince and development of human body, however, it is easily contaminated by microbes and extremes of temperature (Ogbonna *et al*., 2013).

Kunu-aya (non-alcoholic beverage) is locally produced from tigernut and is also called tigernut milk (Ndukwe *et al*., 2023). Tigernut milk (called "Horchata" in Spain) is a non-alcoholic beverage of creamy appearance obtained from tubers of tigernut plant mixed in water containing sugar . The process involves soaking of tigernuts (8hrs), grinding of the nuts , pressing of the composite and mixing with sugar (Sànchez-Zapata *et al*., 2022).

Kunu-aya is more nutritious than kunu-zaki,however,kunu-zaki has higher protein and other nutrients when compared (Ndukwe *et al*., 2023).Tigernut milk is a refreshing non-alcoholic beverage which is typically prepared with tigernut, sugar and water . It can be a substitute for cow milk because of it natural sweet taste.Tigernutmilk can be categorised as natural tigernut milk, pasturized tigernut milk,sterilized tigernut milk and Ultrahigh temperature tigernut milk (Gambo *et al*., 2014). Natural tigernut milk is prepared with the right amount of tigernut, water and sugar.Pasturized tigernut milk is subjected to heat treatment below 72°C without adding additives, it same as the first. Sterilized tigernut milk is obtained after subjecting the milk to

technological process which eliminates or transform totally or partially its starch content and thermally treated after packaging to eliminate microorganisms. UHT tigernut milk is subjected to process that eliminate the starch content and thermally treated at Ultrahigh temperature to eliminate microorganism (Gambo *et al*., 2014).

Tigernut is not a realnut; despite its name, tigernut is a thickened, underground storage organ of plant, botanically, a modified stem that accumulates nutrients to fuel new growth (Sànchez-Zapata *et al*., 2022). Tigernut (*cyperus esculentus lativum*) is a member of the family cyperaceae, order commelinales, class Lilopsida, division Magnoliophyta. Tigernut is an underutilized tuber that produces rhizomes from it base, it almost spherical, it grows freely and consumed widely in Nigeria , parts of west Africa, east Africa ,part of Europe, in Spain where it called chufa . It is known in Nigeria as "Aya" in Hausa , "Ofio" in Yoruba and "Akiansa " in Igbo and grown in various forms black, yellow and brown (Gambo *et al*., 2014). Tigernut and its product tigernut milk had been reported to confer health benefits such as prevention and treatment of colon cancer, diabetes, heart disease,thrombosis, flatulence, indigestion, diarrhea, and dysentery (Afusat, 2022 ;Hayatu,2022; Sànchez-Zapata *et al*.,2022). However, there are reports that tigernuts produces mycotoxin (Sànchez-Zapata *et al*., 2022).*Cyperus esculentus* ,also known as tigernut sedge ,chufa sedge , yellow nut sedge, and earth almond, is a species of sedge that grow in warm, mild to subtropical climates in the Northern Hemisphere (Hayatu,2022). Tigernuts is rich in iron, potassium and phosphorus, however, the principal components are carbohydrates (Hayatu, 2022; Sànchez-Zapata *et al*., 2022).

Cultivation of tigernut started since early times in South Europe and West Africa for its small tuberous rhizomes, which can be eaten raw, roasted dried, baked or made into a refreshing beverage called kunu-Aya. Locally produced drinks are liquids mainly processed from animal or plant sources. They may be regarded as stimulants such as juices, soft drink, or as milk. Locally produced drinks could be classified as either alcoholic or non-alcholic or based on the process involved in preparation can be categorized as industrially processed or traditionally processed beverages (Oyedum and Agbala, 2022). In Nigeria, it is frequently consumed in its raw form, either as a fresh fruit or in its dried alternative, and in other cases, it is crushed and juiced to produce tiger nut milk. Consumption of raw tiger nuts and tiger nut milk has numerous health

benefits; it is a valuable source of vegetable oils (which reduces the risk of heart disease), research has reported the versatility of tiger-nut in Lowering colon cancer risk, Stimulating heart function, Soothing diarrhea symptoms and Reducing inflammation. Tigernut is of various colours yellow, brown and black are grown in Nigeria and they are referred to as “Ofio" in Yoruba, "Akiausa" in Igbo and "Aya" in Hausa. The yellow type is preferred due to its intrinsic qualities, which include its size, beautiful hue, and fleshier structure **Figure 1** . Moreover, it produces more milk after extraction, which contain fewer fats, more proteins, and lower amounts of anti-nutritional compounds, such as polyphenols (Adewunmi *et al*., 2024).

Due to its ability to prevent thrombosis, heart issues, and promote blood circulation, tiger nut milk has been reported to be beneficial for preventing arteriosclerosis (Adewunmi *et al*., 2024). As a result of the high arginine content, which releases the hormone insulin and its optimal combination of sucrose and starch without glucose, tiger nuts without sugar can be used to treat diabetes. The milk is also a suitable beverage for celiac patients who cannot tolerate gluten, and also an option for people with lactose intolerance who require dairy-free substitutes.

However, tigernut milk, frequently encounter a multitude of pathogenic microorganisms — particularly strains exhibiting antibiotic resistance (Oyedum and Agbala, 2022) — throughout the processes of production, handling, and packaging. These pathogenic microorganisms are not solely linked to spoilage but are also associated with foodborne illnesses that may result in significant morbidity and mortality. In light of these hazards, it is imperative to consistently assess the prevalence of resistant microorganisms in widely consumed tigernut milk. Of significant concern is the prevalence of Gram-positive bacteria, including *Staphylococcus aureus*, *Enterococcus faecalis*, and *Bacillus cereus*. These microorganisms are identified as causative agents of foodborne illnesses that can range from benign gastrointestinal disturbances to severe, life-threatening systemic infections. Their capacity to endure in diverse environmental conditions, such as elevated salt concentrations and fluctuating pH levels, renders them persistent contaminants in a variety of food products. This makes it important to assess not just the microbial quality of tigernut milk but also the antibiotic resistance patterns of Gram-positive isolates, with a particular focus on multidrug resistance.

1.2 LITERATURE REVIEW

1.2.1 Origin and distribution Of tigernut

Tigernut is a food that have been consumed since ancient times (Sànchez-Zapata *et al*.,2022).It is commonly consumed food in Egypt. In Egypt, tigernut were roasted and used as sweetmeat(Sànchez-Zapata *et al*., 2022).It is cultivated in ancient mesopotamia between rivers Tigris and Euphrates at that time, Persian and Arab documents, highlighted the nutritive, digestive and disinfective value of tigernut, during this period , tigernut milk was classified as medicinal drink because, it is highly energetic, increase urine production, rich in mineral, majorly phosphorus and potassium and also vitamin C and E (Gambo and Da' u , 2014). Tigernut grows freely, and is widely consumed, use to make drink(kunu-aya) in both East and West Africa countries (Afusat, 2022).However, because production is largely artisanal and often lacks hygienic controls, the drink is vulnerable to microbial contamination (Majumdar *et al*., 2018; Motlhanka *et al*., 2018). Several studies have reported the presence of Gram-positive bacteria such as *Bacillus cereus*, *Staphylococcus* spp., and *Enterococcus* spp. in tigernut-based drinks, some of which exhibit multidrug resistance (Afusat, 2022; Makanjuola *et al*., 2022; Oyedum and Agbala, 2022; Pandukur *et al*., 2019).

Figure1.1: Tigernut

Source: Bado *et al*.,2015

1.2.2. Classification of tigernut

Division: Magnoliophyta

Class: Liliopsida

Order: Commellinales

Family: Cyperaceae

Species: *Cyperus esculentus*

1.2.3. Tigernut Composition

Tigernuts (*Cyperus esculentus*), though popularly referred to as nuts, are underground tubers whose nutritional composition aligns more closely with other root and tuber crops. They are recognised for their high starch content, dietary fibre, and carbohydrates, alongside appreciable levels of sucrose (17.4–20.0%), lipids (≈25.5%), and proteins (≈8.0%). In addition, tigernuts are rich in essential minerals such as calcium, sodium, phosphorus, potassium, magnesium, zinc, and iron, with trace amounts of copper, as well as vitamins C and E. This combination of energy-yielding macronutrients with micronutrients enhances their dietary and health-promoting potential (Afusat, 2022).Comparative studies have shown that tigernut tubers contain nearly twice the starch content of potatoes and sweet potatoes, and they also provide notable levels of total sugars, reducing sugars, and sucrose (Gambo and Da'u, 2014). Although their sugar concentration is relatively lower than that of some other tuber and nut crops, the characteristic sweet and nutty flavour of tigernuts—largely attributed to their natural sugars—has contributed to their popularity as a snack and their growing utilisation in food processing and product development (Afusat, 2022; Ogbonna *et al*., 2013).In terms of mineral composition, tigernuts supply substantial amounts of calcium, sodium, and phosphorus, but comparatively lower levels of magnesium, manganese, iron, zinc, and copper (Gambo and Da'u, 2014). The calcium content is sufficient to support bone and dental health in infants, while iron plays a vital role in haemoglobin synthesis and blood formation. Based on these nutritional attributes, tigernut flour has been recommended as a fortifying ingredient to improve the mineral profile of cereal-based flours and complementary foods (Oladele and Aina, 2007; Afusat, 2022).

1.2.4 Products obtained from tigernut

The principal product of tigernut is tigernut milk, commonly referred to in Nigeria as kunu-aya. Traditionally, kunu is prepared using cereals such as millet or sorghum, to which spices (e.g., ginger, alligator pepper, dandelion, licorice) and sugar are added during processing. Due to its superior nutritional composition, tigernut has been recognised as a suitable substitute for these cereals in the preparation of kunu-aya (Belewu and Abodunrin, 2008; Sànchez-Zapata *et al*., 2022). Beyond Nigeria, the most renowned tigernut-based beverage is "horchata de chufa," widely consumed in Spain, which highlights the versatility of tigernut as a raw material in beverage technology (Coşkuner *et al*., 2002). In addition to beverage production, tigernut has been utilised as a flavouring agent in ice cream, while roasted tigernut flour has been incorporated into biscuits, bakery products, and other processed foods. Tigernut is also used industrially in the production of oil, soap, and starch extracts (Adejuyitan, 2011).Regional variations in tigernut beverages further demonstrate its cultural and nutritional significance. For instance, in Ghana, beverages are produced either from roasted or unroasted tigernuts, reflecting diverse preparation methods and consumer preferences (Sanful, 2009).Apart from beverages, tigernut is incorporated into dakuwa, a popular snack in northern Nigeria that is also consumed in other parts of the country. Traditionally, dakuwa is prepared from maize, groundnut, sugar, and spices; however, tigernut may replace maize either partially or wholly in the formulation. The preparation involves roasting the ingredients to develop the characteristic flavour, colour, and texture of the product. According to Oladele *et al*. (2009), consumer acceptance of dakuwa is largely influenced by sensory properties, which are primarily determined by the roasting process. Similarly, Ocheme *et al*. (2013) emphasised that dakuwa production is largely artisanal, with no standardised processing techniques. Oladele *et al*. (2013) further investigated the effect of roasting temperatures (150°C, 160°C, and 170°C) on the sensory quality of tigernut-based dakuwa. Their study showed that roasting tigernut at 150°C yielded a product most acceptable in terms of colour, texture, and flavour, while roasting at 160°C enhanced overall taste acceptability.

1.2.5 Preparation of tigernut milk (locally)

The production process involves washing of the tigernuts,followed by soaking of the tigernuts in water for 2 - 8 hrs ,afterwards,the soaked tigernut is grinded and mixed in water in the ratio

3L/kg. The mixture is allowed to soften, then sugar is added, followed by seiving to obtain tigernut milk (Afusat, 2022).

Fresh Tigernuts
|
Sorting and Washing
|
Grinding/milling
|
Extracting/Sieving
|
Pasteurizing (72°C for 5 sec)
|
Homogenizing
|
Bottling
|
Cooling
|
Tigernut milk

Figure1.2: Flow chart for traditional preparation of Tigernut milk

Source: Gambo and Da' u , 2014.

Regardless of , the nutritional content of tigernut milk such as it refreshing property, its nutritional composition, and health benefits, it is produce in rural areas by people with poor hygiene practices as a result this locally produced non-alcoholic beverage is exposed to certain resistant pathogenic bacteria during the production and packaging processes (Oyedum *et al*., 2022 ; Pandukur *et al*., 2019).

1.2.6 Sources of microorganism in tigernut milk

Foodborne bacterial contamination remains a pervasive issue in traditional and street-vended foods, often linked to unsafe handling and processing conditions (Majumdar *et al*., 2018; Petruzzi *et al*., 2017). Tigernut beverages like kunu-aya are no exception, with several studies documenting diverse bacterial populations inhabiting these drinks. For example, Afusat (2022) isolated various bacteria from tigernut drinks in Lagos State, highlighting significant microbial diversity that includes potential pathogens. Makanjuola *et al*. (2022) similarly reported the presence of bacteria in kunu-aya sold in Kaduna State, emphasizing hygiene gaps in production and sale which can facilitate contamination.

More critically, pathogens such as *Bacillus cereus*, known for causing foodborne illnesses, have been isolated from various food matrices including dairy and meat (Abraha *et al*., 2017; Awadelkareem, 2006; Ramarao *et al*., 2020) and have been detected in tigernut products (Oyedum and Agbala 2022).. The detection of these pathogens signals a substantial risk to public health, especially in the absence of stringent microbial controls. Maduka *et al*. (2022) and Nwachukwu (2024) conducted microbiological and nutritional quality assessments of tigernut milk products across different Nigerian cities, revealing variations in microbial loads that reflect inconsistencies in production, storage, and handling practices.

While microbial contamination remains a concern, some investigations (Nodu *et al*., 2021) indicate the coexistence of probiotic bacterial strains within tigernut drinks, which may confer health benefits. Nevertheless, the potential for pathogenic contamination remains predominant, requiring targeted monitoring.

1.2.7 Gram-Positive bacteria isolated from tigernut milk

Research on tigernut drinks across Nigeria have consistently identifies Gram-positive contaminants. *Bacillus cereus*, a spore-forming pathogen associated with emetic and diarrhoeal syndromes, has been isolated from cow milk and related beverages, indicating its potential presence in tigernut milk (Abraha *et al*., 2017; Ramarao *et al*., 2020). Likewise, *Staphylococcu*s and *Enterococcus* species have frequently been recovered from street-vended tigernut drinks (Afusat, 2022; Makanjuola *et al*., 2022; Nodu *et al*., 2021). These organisms are often introduced through raw ingredients, contaminated water, or unhygienic handling practices (Petruzzi *et al*., 2017).

1.2.8 *Bacillus cereus*

Bacillus cereus, a member of the family Bacillaceae, is a Gram-positive, motile, spore-forming rod widely distributed in the environment, particularly in soil, fresh water, and marine habitats (Iyiola, 2022). The spores of *B. cereus* possess appendages and pili, which increase their hydrophobicity compared to other *Bacillus* spores. This characteristic enhances their adherence to various surfaces and contributes to their survival, even after cleaning and sanitation procedures. Vegetative cells of the organism are capable of growth over a wide temperature range (4–55 °C), with the optimal range typically between 30 and 40 °C depending on the strain (Abraha *et al*., 2017)..The organism can grow within a pH range of 4.9–9.3, although low pH tends to inhibit growth; however, in complex food matrices such as meat, limited growth has been observed at pH 4.35 (Iyiola, 2022).

In addition, *B. cereus* tolerates reduced water activity (aw), with a minimum value of 0.93 reported for growth. Some studies suggest an even lower threshold of 0.912, consistent with the observation that foods such as fried rice, with aw values ranging between 0.912 and 0.961, readily support its proliferation (Iyiola, 2022). The organism is an important foodborne pathogen associated with two syndromes: the emetic and diarrheal types of illness. The emetic syndrome results from a thermostable toxin synthesized in food prior to ingestion, while the diarrheal form arises from thermolabile enterotoxins produced in the small intestine during bacterial growth. Clinically, the emetic syndrome is characterized by nausea and vomiting within 1–5 h of consumption, with symptoms persisting for 6–24 h, whereas the diarrheal syndrome presents after a longer incubation period of 8–16 h, with abdominal pain, watery diarrhea, and occasional nausea, typically lasting 12–24 h (Iyiola, 2022).

1.2.9. *Staphylococcus aureus*

Staphylococcus aureus is a Gram-positive, non-motile coccus that may occur singly, in pairs, tetrads, short chains, or in the characteristic grape-like clusters (Iyiola, 2022). Its growth parameters are strain-dependent, with reported growth between 7–47.8 °C and an optimum range of 40–45 °C. Production of enterotoxins generally occurs within 10–46 °C. The organism also thrives over a broad pH spectrum (4.5–9.3), with optimal growth close to neutrality (pH 7.0–7.5), and displays notable halotolerance, being able to multiply in environments containing more than

10% NaCl (Iyiola, 2022).Outside host environments, *S. aureus* exhibits remarkable resilience, surviving for extended periods under desiccated conditions. It has been isolated from air, dust, sewage, and water, underscoring its position as one of the most resistant non-spore-forming pathogens (Iyiola, 2022). The presence of *S. aureus* in food poses significant public health risks due to its ability to produce heat-stable enterotoxins that cause staphylococcal food poisoning. Among the nine identified enterotoxins (A, B, C1, C2, C3, D, E, F, and G), types A and D are most frequently implicated in outbreaks (Iyiola, 2022).Importantly, these toxins display exceptional thermal stability, often requiring higher temperatures for inactivation than those sufficient to kill the vegetative cells (Mbachu *et al*., 2014). For example, enterotoxin A is more heat-sensitive compared to toxins B and C, requiring heating at 80 °C for 180 s or at 100 °C for 60 s to lose serological activity (Mbachu *et al*., 2014).

1.3.0. Enterococci

Enterococci are Gram-positive cocci that occur singly, in pairs, or in short chains. They are facultatively anaerobic, non-spore-forming, and motile in certain species. Fermentation of glucose by these organisms typically results in the production of lactic acid (Awadelkareem, 2006). According to Sherman (1937), the designation “Enterococci” should apply specifically to streptococci capable of growth at both 10 °C and 45 °C, in environments containing 6.5% NaCl, and at pH 9.6, as well as to those able to survive exposure to 60 °C for 30 minutes. Additionally, enterococci characteristically hydrolyze esculin in the presence of bile, are catalase- and oxidase-negative, and produce pyrrolidonyl arylamidase (Awadelkareem, 2006).Although not regarded as primary foodborne pathogens, *Enterococcus* species may contribute to disease through several mechanisms. They possess virulence factors including adhesins, invasins, hemolysins, cytolysins, and gelatinase, which enhance their ability to cause opportunistic infections, particularly in immunocompromised individuals (Fisher and Phillips, 2009;Hollenbeck and Rice,2012). A significant public health concern is their role as reservoirs of antimicrobial resistance. Multidrug-resistant (MDR) strains, including vancomycin-resistant enterococci (VRE), have been isolated from food sources, raising the possibility of horizontal gene transfer to other pathogenic bacteria within the food chain (Fisher and Phillips, 2009).Furthermore, some enterococcal strains act as spoilage organisms, negatively affecting food quality.

1.3 .1 Factors Contributing to Microbial growth in food

The growth and multiplication of microorganisms in food are largely dependent on the surrounding food environment. Several factors influence the ability of bacteria to persist and proliferate in both raw and processed foods. These include intrinsic properties of the food, extrinsic environmental conditions, and implicit factors, along with the effects of different processing techniques. Such factors not only affect microbial survival and growth but also determine the metabolic pathways used by microorganisms for energy production and waste formation. Consequently, the likelihood of contamination with significant microorganisms—together with the impact of processing—and the risk of further growth or toxin production, establish the need for strict time and temperature control measures.

Intrinsic Factor

Intrinsic factors refer to the inherent properties of food that influence the growth and survival of microorganisms. Among these, water activity, food structure, nutrient composition, and pH play critical roles in determining microbial proliferation and, consequently, the shelf life of food products (Petruzzi *et al*., 2017).

Water Activity (aw):

Water activity is a key determinant of microbial growth, as microorganisms require a moist environment for metabolic processes. A reduction in water activity—through drying, addition of solutes such as salt or sugar, or other preservation methods—limits microbial growth and enhances product stability, thereby extending shelf life (Petruzzi *et al*., 2017).

Food Structure:

The physical structure of food also provides a natural defense against microbial invasion. For example, protective layers such as fruit peels, nut shells, and eggshells reduce microbial penetration, though microorganisms often colonize the food surface. The rate of spoilage also depends on consistency: liquid foods are highly perishable because microorganisms can disperse rapidly, semi-solid foods similarly support quick spoilage, whereas solid foods generally deteriorate from the surface inward (Petruzzi *et al*., 2017).

Food Composition:

The nutrient composition of food dictates the type of spoilage organisms that predominate. Carbohydrate-rich foods are more prone to fermentation by saccharolytic microorganisms, protein-rich foods are susceptible to degradation by proteolytic species, while lipid-containing foods are vulnerable to spoilage by lipolytic organisms (Petruzzi *et al*., 2017).

pH:

The hydrogen ion concentration of food strongly influences the spectrum of microorganisms able to grow. Acidic foods tend to favor the growth of yeasts and molds, whereas bacterial growth is generally enhanced in neutral to alkaline environments (Petruzzi *et al*., 2017).

Extrinsic Factors

Extrinsic factors are environmental conditions surrounding food during processing, storage, and distribution, which significantly influence microbial growth and spoilage. Among the most important extrinsic factors are temperature and gaseous conditions (Petruzzi *et al*., 2017).

Temperature:

Temperature plays a pivotal role in determining microbial proliferation and the rate of food spoilage. Microorganisms are broadly classified into three groups based on their temperature tolerance: psychrophiles, mesophiles, and thermophiles. Psychrophilic species, such as *Yersinia enterocolitica,* thrive at low temperatures, with a growth range of –5 °C to 35 °C. Mesophiles, have a temperature of 5-15℃ minimum range ,optimum range 30-45℃ and maximum 35-37℃ (Petruzzi *et al.*,2017).

Gasoeus conditions:

Spoilage by aerobic organisms occur at surface of foods , these foods have low oxidation-reduction potential at the surface only while anaerobic organism grow in the food (Petruzzi e*t al.*,2017).

1.3.2 Antibiotic Resistance in Foodborne Gram-Positive Bacteria

Antimicrobial resistance (AMR) among foodborne bacteria constitutes a recognized global concern, with the food supply chain being a major significant pathway for transmission (Capita and Alonso-Calleja, 2013). Established definitions for multidrug-resistant (MDR), extensively drug-resistant (XDR), and pan-drug-resistant (PDR) bacteria offer a systematic approach for delineating resistance phenotypes (Magiorakos *et al*., 2012). Investigations into tigernut beverages have identified Gram-positive isolates exhibiting resistance to various classes of antibiotics, including β-lactams, tetracyclines, and aminoglycosides (Pandukur *et al*., 2019; Oyedum and Agbala, 2022). Such results correspond with extensive evidence indi35 cating the emergence of resistance influenced by the misuse of antibiotics in both humans and animals, horizontal gene transfer, and inadequate food-handling methodologies (Aminov, 2010; Capita and Alonso-Calleja, 2013).

1.3.3. Public Health Implications

The identification of multidrug-resistant Gram-positive bacteria in tigernut milk presents considerable concerns regarding food safety. *Bacillus cereus* is capable of inducing acute intoxications, whereas resistant strains of *Staphylococcus* and *Enterococcus* present threats of opportunistic infections that may be unresponsive to first-line therapeutic interventions (Abraha *et al*., 2017; Capita and Alonso-Calleja, 2013). Additionally, contaminated beverages may serve as reservoirs for resistance genes that can be transferred to commensal gut microbiota or other pathogenic organisms (Aminov, 2010). These observations highlight the dual threats of foodborne illnesses and the propagation of antimicrobial resistance through a beverage that is widely consumed.

1.4. Antibiotic Resistant Gram Positive Bacteria in Tigernut Milk

Tigernut is acknowledged as a tuber rich in nutrients, encompassing significant amounts of fiber, lipids, sugars, and a moderate level of protein, alongside minerals like potassium, phosphorus, and calcium, as well as vitamins E and C (Gambo and Da'u, 2014; Sánchez-Zapata *et al*., 2022). These components contribute to its growing popularity in beverages such as kunu-aya (horchata) and its noted health benefits, especially concerning metabolic and digestive wellness (Ogbonna *et al*., 2013). Concurrently, tigernut has been linked to contamination risks, including the

presence of mycotoxins in tubers and nuts sold by street vendors (Adewunmi *et al*., 2024). In order to ensure consumer safety, various processing methods—such as pasteurization, sterilization, and ultra-high temperature treatments—have been utilized to maintain sensory quality while minimizing microbial threats (Petruzzi *et al*., 2017). Findings from literatures consistently shows that kunu-aya made in unsanitary conditions is susceptible to microbial contamination at several stages of processing, which includes washing, soaking, grinding, extraction, and storage. Contamination sources comprise water, equipment, handlers, and packaging (Afusat, 2022; Iyiola, 2022; Makanjuola *et al*., 2022).These findings highlight systemic lapses in hygienic practices during small-scale processing and retailing of tigernut milk (Majumdar *et al*., 2018).Gram-positive bacteria have been repeatedly implicated in foodborne outbreaks linked to beverages. *S. aureus*, commonly introduced via handlers, is particularly concerning because of its ability to produce heat-stable enterotoxins (Majumdar *et al*., 2018). *E. faecalis*, often originating from contaminated water or equipment, displays environmental resilience and salt tolerance (Awadelkareem, 2006). Similarly, *B. cereus* can survive mild heat treatments through its spores and is capable of producing diarrhoeal and emetic toxins, while thriving across a broad range of pH and temperature conditions (Abraha *et al*., 2017; Ramarao *et al*., 2020).In many developing regions, selective and differential media such as Mannitol Salt Agar, Bile Esculin Agar, and *Bacillus cereus* Agar, alongside conventional biochemical tests, remain the standard approach to identification (Al-joda and Jasu, 2021).The emergence of antimicrobial resistance in foodborne bacteria represents a significant global challenge. Food-handling environments provide opportunities for resistant strains to persist and spread, while ready-to-drink beverages may act as transmission vehicles (Capita and Alonso-Calleja, 2013). High resistance to β-lactams and macrolides has been documented among Gram-positive isolates, reflecting long-standing antibiotic usage, while fluoroquinolones have shown comparatively better activity, though declining susceptibility has also been reported (Oyedum and Agbala, 2022).To ensure comparability across studies, consistent definitions—such as the international MDR classification proposed by Magiorakos *et al*. (2012).Across existing research, several patterns emerge. Gram-positive isolates (*S. aureus, E. faecalis, B. cereus*) are frequently detected in non-industrial tigernut beverages (Afusat, 2022; Makanjuola *et al*., 2022; Oyedum and Agbala, 2022). These isolates commonly display resistance to widely available antibiotics, including

amoxicillin, erythromycin, and cephalosporins, while resistance to fluoroquinolones, though lower, remains a concern (Pandukur *et al*., 2019). Moreover, weaknesses in processing and storage practices—such as inadequate pasteurization, poor refrigeration, and post-processing contamination—create conditions conducive to microbial growth and toxin production (Majumdar *et al*., 2018; Petruzzi *et al*., 2017).These recurring findings underline the potential of tigernut milk as both a vehicle for antimicrobial resistance(AMR) transmission and a source of foodborne illness, reinforcing the need for improved hygienic standards and monitoring (Capita and Alonso-Calleja, 2013).Review of prior studies provides methodological justification for the present research. Bacterial confirmation through colonial and biochemical test is a common practice in food safety studies (Afusat, 2022; Iyiola, 2022). In addition, disc diffusion testing against commonly available antibiotics remains a practical method, while allowing comparability with earlier regional surveys (Pandukur *et al*., 2019; Oyedum and Agbala, 2022).Although tigernut milk is widely consumed, existing studies often treat branded and unbranded products separately, few studies provide data on species level identification, existing research often focuses on isolated samples or lacks detailed antibiograms.

1.5 STATEMENT OF PROBLEM

The proliferation of multidrug resistant Gram-positive bacteria within food products constitutes a significant public health concern, as these pathogens may lead to infections that are challenging to manage therapeutically. Notwithstanding the increasing popularity of tigernut milk, there exists a paucity of research investigating the prevalence of antibiotic-resistant Gram-positive bacteria in this particular beverage.

1.6 SIGNIFICANCE OF STUDY

This study provides data on the occurrence, prevalence, and antibiotic resistance patterns of Gram-positive bacteria in tigernut milk, with a focus on multidrug resistance. The findings can inform public health policies, food safety regulations, and community-level hygiene practices to reduce the risk of AMR transmission through locally produced beverages.

1.7 AIM OF STUDY

To isolate and identify Gram-positive bacteria from tigernut milk and determine their antibiotic resistance profiles, with emphasis on multidrug resistance.

1.8 OBJECTIVES OF STUDY

To isolate and identify Gram-positive bacteria present from branded and unbranded tigernut milk

To determine the occurrence and prevalence of each Gram-positive bacterium.

To determine the antibiogram of the isolates.

CHAPTER TWO

2.1 MATERIALS AND METHODS

2.1.1 Equipment and Apparatus used

The following materials were used for the experiment; test-tubes, test-tube rack, syringes,petri-dishes, masking-tape, glass spreader,alcohol, cotton wool, foil paper, marker, microscope, inoculating loop, Bunsen burner, autoclave, measuring cylinder, incubator, glass slides, coverslips, weighing machine, distilled water, saline water,hand gloves, heating block, refrigerator, spirit lamp, beakers, conical flask, spectrophotometer ,and antibiotics discs.

2.1.2 Reagents and Media used

Nutrient agar, Mueller Hinton agar,Citrate media, Mannitol salt agar, Gram's iodine solution, crystal violet, Lugol's iodine, safranin, peptone ,oil immersion, 3% hydrogen peroxide, filter paper, blood plasma, Bile esculin agar, *Bacillus cereus* agar.

2.2 Sterilization of materials

All glass wares including beakers, conical flasks, measuring cylinder were washed, rinsed with clean water, dried and autoclaved for 15 minutes at 121°C .All media were prepared following manufacturer's instructions. The media were prepared in a conical flask, plugged with cotton wool wrapped in foil paper and sterilized for 15 minutes at 121°C by autoclaving, afterwards, media was aseptically dispensed onto sterile petri-dishes. Inoculating loop was sterilized by heating to red hot in the spirit lamp.

2.3 Sample collection

A total of twenty-five tigernut milk samples were obtained randomly from selected areas in Agege, Abulegba, Unilag and Bariga, in quintuple. Five samples of unbranded tigernut milk were obtained from Agege and Unilag respectively. Five samples of branded tigernut milk were obtained from Abulegba while ten samples of branded tigernut milk were obtained from Bariga. The samples were assigned different codes for the branded and unbranded tigernut milk within this location. The samples were labelled properly, placed in a polythene bag and taken to the microbiology laboratory of department of microbiology.

2.4 Microbiological analysis

2.4.1 Media preparation

Media was prepared following manufacturer's instruction(s).

2.4.2 Serial dilution

Samples collected were first pre-enriched .1ml of samples were dispensed in 9ml of peptone water, incubated at 37°C for 24 hours, afterwards, five-fold serial dilution was carried out by dispensing 1mL of each sample of tigernut milk which serves as stock culture into a sterile test tube containing 9mL of sterile saline water using sterile pipette tips, and test tube was mixed and labelled 10^{-1}. 1mL of dilution (10^{-1}) was then introduced into another test tube (10^{-2}) containing 9mL of saline water, the sample was serially diluted to the required dilution 10^{-5}. 0.1mL of dilution 10^{-5} was inoculated onto labelled sterile petri-dishes in duplicates.

2.4.3 Plating methods

Inoculation was carried out using spread plate method. Aliquot of 0.1mL of inoculum was aseptically pipetted from dilution 10^{-5} and dispensed onto bile esculin agar, *Bacillus cereus* agar and Mannitol salt agar, then ,the inoculum was spread on the agar surface with sterile glass spreader, plates were incubated at 37°C for 24 hours to obtain a bacterial count in CFU/mL.

2.5 Characterization and identification of bacterial isolates

2.5.1 Colonial Morphology

Colony morphology is the characters of a bacterial colony on the agar plate that is seen with the naked eye. Observing colony morphology is an important skill in microbiology laboratory to identify microorganisms. The shape, size, colour, elevation ,margin ,opacity, texture, and change in medium were examined on Mannitol salt agar (MSA), *Bacillus cereus* agar base, Bile esculin azide agar after incubation for 48 hours at 37°C.Colonies with required features were subcultured to obtain pure isolates, the pure isolates were preserved on agar slant at 4℃.

2.6 Identification and Biochemical test of bacterial Isolates

2.6.1 Gram's stain

Sterile inoculating loop was used to pick a colony on agar plate onto clean glass slides. The inoculating loop is use to make a smear on the glass slides and heat fixed, the slides were labelled and arranged on a rack , afterwards, each slide was stained with crystal violet for 30-60

seconds. The dye was drained and fixed with Lugol's iodine for 30 seconds. The slides were rinsed with water, decolorized with 95% alcohol for 10-15 seconds, and rinsed with tap water. The slides were counter stained with safranin for 30 seconds, and then rinsed, air dried and examined under the microscope using oil immersion (x100). Gram positive bacteria stained purple.

2.6.2 Catalase test

A suspension of 18-24 hours old culture of the test organisms was made on clean glass slides by mixing with 3% of hydrogen peroxide. The evolution of bubbles indicated the presence of catalase enzyme. That is, organisms can breakdown hydrogen and oxygen.

2.6.3 Coagulase test

A suspension of the test organisms were made on clean glass slides by mixing with blood plasma. The presence of agglutination indicated a positive reaction.

2.6.4 Motility test

Motility was tested by stabbing the isolated bacteria using straight wire loop into a semi solid medium. The medium was then incubated for 24 hours at 37°C. The growth of non-motile organism was confined to the stab line, while the motile one radiated away from the stab line.

2.6.5 Citrate test

Simmon’s citrate medium is a nutrient substrate that offers ammonium salts as the only source of nitrogen and citrate as the only carbon source. The degradation of citrate leads to alkalinisation of the medium which is indicated by the pH indicator bromothymol blue changing colour from green to deep blue. Slants of simmon’s citrate agar plates were inoculated with the isolates and incubated at 35°C for 24 - 48 hrs. Colour change from green to blue indicated a positive result.

2.6.5 Salt tolerance test

Colonies were inoculated into peptone water containing 6.5% Nacl. The inoculated broth was incubated at 37°C for 24 hours .Growth was indicated by development of turbidity which was measured using a spectrophotometer with an absorbance of 620nm.

2.7 Antimicrobial susceptibility test

Thirty-eight grams (38g) of Mueller Hinton agar was weighed into 1L of distilled water in a conical flask, stirred, to dissolve and subsequently sterilized at 121°C for 15 minutes in an autoclave. The Mueller Hinton agar was poured into sterile petri-dishes,the plates were allowed to solidify. Suspension of the organisms were made by inoculating the test organisms of 24 hours into 1ml of water and compared to McFarland turbidity standard. Afterwards, 0.1mL aliquot of test organism suspension was pipetted on the agar surface and spread aseptically using glass spreader. The discs containing antibiotics was placed on the inoculated plates. The plates were incubated at 37°C for 24 hours.

After incubation, the culture plates were examined for area of no growth around the discs and measured in millimeters with a ruler. Bacterial strain resistant to antimicrobial grow to the discs edge while sensitive strains are inhibited at a distance from the disc. The zone of inhibition (diameters) around the antibiotic discs were categorized as sensitive (S), resistant (R) and intermediate (I) based on Clinical and Laboratory Standards Institute (CLSI, 2024). Disc containing Azithromycin, Erythromycin, ciprofloxacin, amoxicillin, levofloxacin, ofloxacin, ceftriazone, tetracycline, pefloxacin and cotrimazole were used.

Statistical analysis

Statistical analysis was carried out using IBM Statistical Package for Social Sciences(SPSS) version 27. In this study, statistical significance was determined using an Independent Samples t-test at a 95% confidence level ($p < 0.05$). The results showed no statistically significant differences ($p > 0.05$) in bacterial loads between branded and unbranded tigernut milk samples across all selective media.

CHAPTER THREE
RESULTS

Tigernut milk, a cereal-based product widely consumed in Nigeria has been discovered to be a source of transmission of multidrug resistant Gram-positive microorganisms. This study therefore using standard microbiological methods isolated and identified the Gram-positive bacteria present in tigernut milk and studied their resistance to antibiotics. The bacteria load of the tigernut milk samples were high and ranged from 0 CFU/mL to 2.2 x 10^8 CFU/mL. Overall, samples TM^2BA, TE^4UA and TC^2BA had very high microbial loads 1.225 x10^8,1.5 x10^8 and 2.2 x 10 ^{8} CFU /mL respectively , while samples TM^5BB, TE^2BA, TE^5BA ,TC^5BA and TC^1BB had no growth **Appendix I**. Sample TM^5BB, TE^2BA, TE^5BA, TC^5BA and TC^1BB which were branded had no growth. In comparison of the total bacterial count, branded sample had the highest bacterial load **Appendix I**. The morphological characteristics of isolates from tigernut milk were shown in **Table 3.1.** All isolates were gram positive. The biochemical characteristics of the isolates showed most of the isolates were catalase positive,salt tolerance positive and citrate positive, seven of the isolates were non-motile while thirteen were motile, seven were coagulase positive, while thirteen were coagulase negative **Table 3.2.** The bacterial isolates were either susceptible, intermediate, or resistant to all tested antibiotics. Most isolates were resistant to Erythromycin (80%), cotrimazole (70%) and Amoxicillin (75%) while majority of the isolates were resistant to Ceftriazone (95%) **Figure 3.5.**.The most occuring species are *Staphylococcus aureus* and *Enterococcus faecalis* occuring at 35%,with *Bacillus cereus* occuring 30% **Figure 3.4.** Out of 20 isolates,17 isolates (85%) are resistant to at least three different classes(≥3) of antibiotics, which qualifies as Multidrug Resistant (MDR) under the standard definition (resistance to at least 3 antimicrobial classes).Only 3 isolates were resistant to fewer than 3 antibiotics.The most common resistance level was 4 antibiotics , followed by 3 antibiotics .The distribution suggests a high prevalence of MDR, indicating a serious antibiotic resistance concern**.**A total of 20 isolates were identified, seven *Staphylococcus aureus*, seven *Enterococcus faecalis* and six *Bacillus cereus* .Statistical analysis was conducted to compare the microbial load between branded and unbranded tigernut milk. The results showed no significant difference in bacterial load between the two sample types , p value ranged from 0.2-0.8,indicating that branding does not necessarily equate to microbiological safety **Appendix II.**

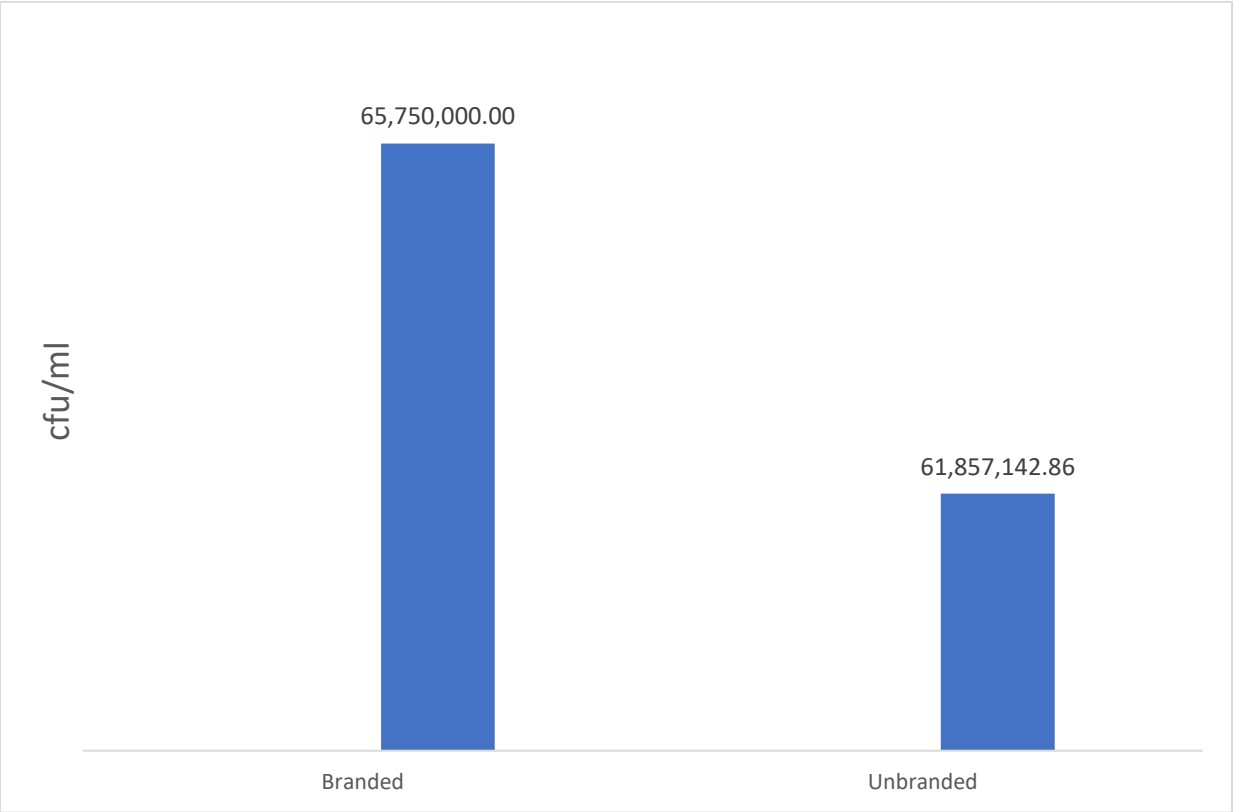

Figure 3.1: Comparison of bacterial load in branded and unbranded tigernut milk on Manittol Salt Agar

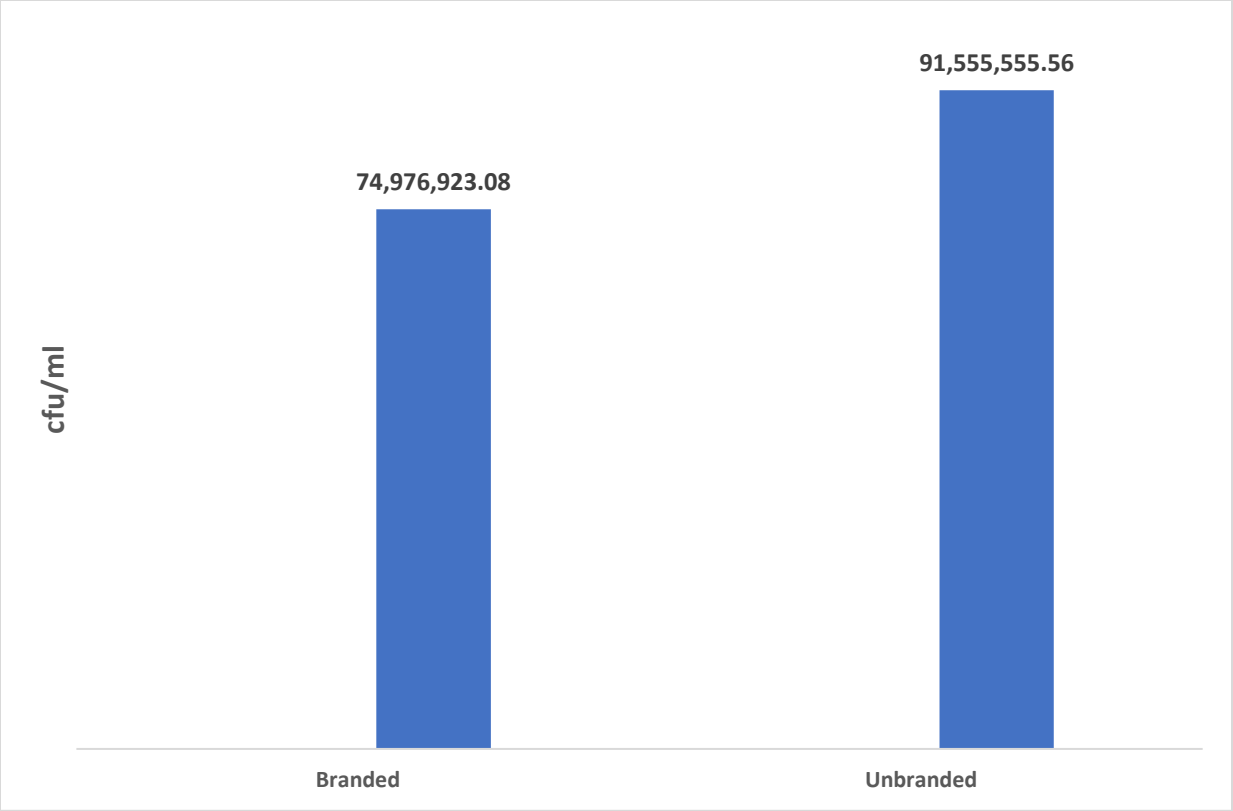

Figure 3.2: Comparison of bacterial load in branded and unbranded tigernut milk on Bile Esculin Agar

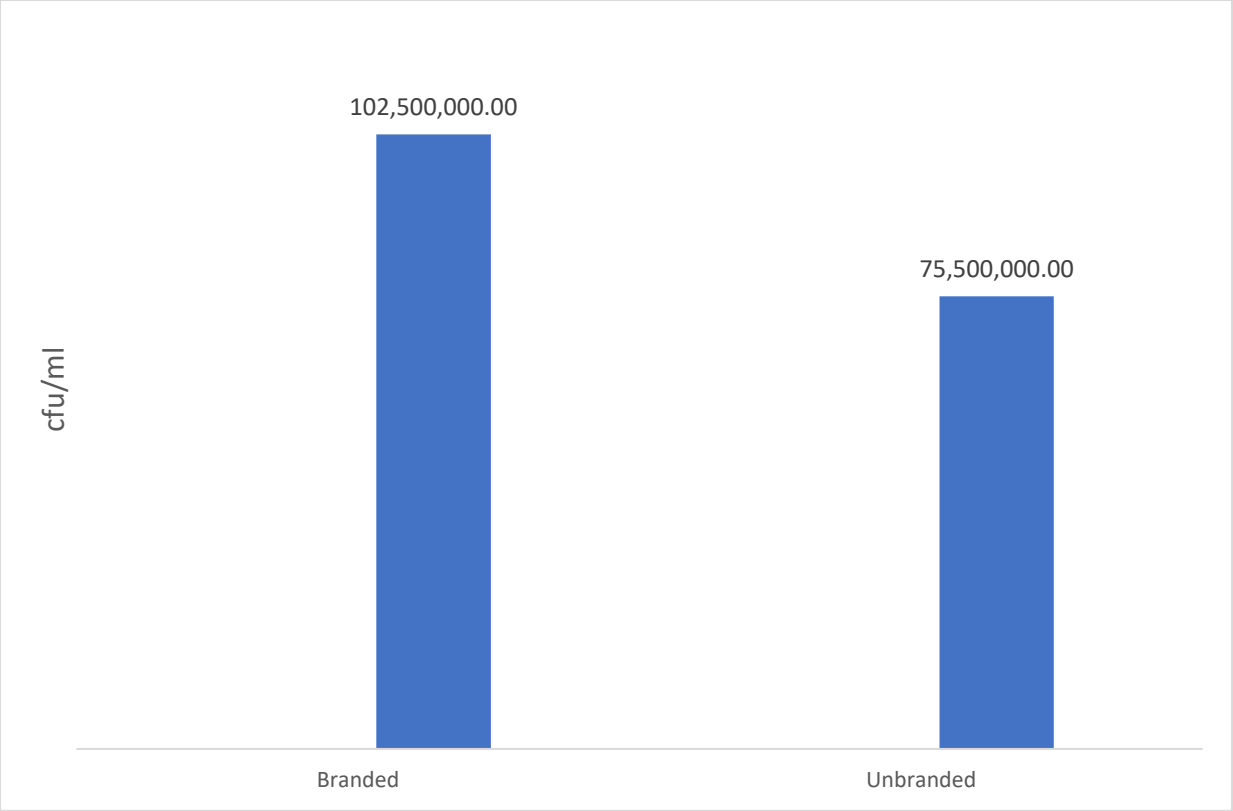

Figure 3.3: Comparison of bacterial load in branded and unbranded tigernut milk on Bacillus cereaus agar

Table 3. 1: **Cultural characteristics of bacterial colonies from tigernut milk samples**

Code	Shape	Size	Colour	Elevation	Margin	Change in medium	Opacity	Texture
TM¹UA	Round	Small	Yellow	Raised	Entire	Yellow	Opaque	Mucoid
TM²UA	Round	Small	Yellow	Raised	Entire	Yellow	Opaque	Mucoid
TM³UA	Round	Small	Yellow	Raised	Entire	Yellow	Opaque	Mucoid
TM⁴UA	Round	Small	Yellow	Raised	Entire	Yellow	Opaque	Mucoid
TM⁵UA	Round	Small	Yellow	Raised	Entire	Yellow	Opaque	Mucoid
TM¹UB	Round	Small	Yellow	Raised	Entire	Yellow	Opaque	Mucoid
TM²UB	Round	Small	Yellow	Raised	Entire	Yellow	Opaque	Mucoid
TE¹UA	Round	Large	Black	Raised	Entire	Black	Opaque	Mucoid
TE²UA	Round	Large	Black	Raised	Entire	Black	Opaque	Mucoid
TE³UA	Round	Large	Black	Raised	Entire	Black	Opaque	Mucoid
TE⁴UA	Round	Large	Black	Raised	Entire	Black	Opaque	Mucoid
TE⁵UA	Round	Large	Black	Raised	Entire	Black	Opaque	Mucoid
TE¹UB	Round	Large	Black	Raised	Entire	Black	Opaque	Mucoid
TE²UB	Round	Large	Black	Raised	Entire	Black	Opaque	Mucoid
TC¹UA	Round	Moderate	Pink	Raised	Entire	Pink	Opaque	Mucoid
TC²UA	Round	Moderate	Pink	Raised	Entire	Pink	Opaque	Mucoid
TC³UA	Round	Moderate	Pink	Raised	Entire	Pink	Opaque	Mucoid
TC⁴UA	Round	Moderate	Pink	Raised	Entire	Pink	Opaque	Mucoid
TC⁵UA	Round	Moderate	Pink	Raised	Entire	Pink	Opaque	Mucoid
TC¹UB	Round	Moderate	Pink	Raised	Entire	Pink	Opaque	Mucoid

Key: U= unbranded tigernut milk

Table 3.2: Morphiological and biochemical characteristics of bacterial isolates from tigernut milk samples

Code	Gram stain	Shape	Coagulase	Catalase	Motility	Salt tolerance	Citrate	Suspected organisms
TM¹UA	+	Cocci	+	+	Non-motile	+	+	*Staphylococcus aureus*
TM²UA	+	Cocci	+	+	Non-motile	+	+	*Staphylococcus aureus*
TM³UA	+	Cocci	+	+	Non-motile	+	+	*Staphylococcus aureus*
TM⁴UA	+	Cocci	+	+	Non-motile	+	+	*Staphylococcus aureus*
TM⁵UA	+	Cocci	+	+	Non-motile	+	+	*Staphylococcus aureus*
TM¹UB	+	Cocci	+	+	Non-motile	+	+	*Staphylococcus aureus*
TM²UB	+	Cocci	+	+	Non-motile	+	+	*Staphylococcus aureus*
TE¹UA	+	Cocci	-	+	Motile	+	+	*Enterococcus faecalis*
TE²UA	+	Cocci	-	+	Motile	+	+	*Enterococcus faecalis*
TE³UA	+	Cocci	-	+	Motile	+	+	*Enterococcus faecalis*
TE⁴UA	+	Cocci	-	+	Motile	+	+	*Enterococcus faecalis*
TE⁵UA	+	Cocci	-	+	Motile	+	+	*Enterococcus faecalis*
TE¹UB	+	Cocci	-	+	Motile	+	+	*Enterococcus faecalis*
TE²UB	+	Cocci	-	+	Motile	+	+	*Enterococcus faecalis*
TC¹UA	+	Rods	-	+	Motile	+	+	*Bacillus cereus*
TC²UA	+	Rods	-	+	Motile	+	+	*Bacillus cereus*
TC³UA	+	Rods	-	+	Motile	+	+	*Bacillus cereus*
TC⁴UA	+	Rods	-	+	Motile	+	+	*Bacillus cereus*
TC⁵UA	+	Rods	-	+	Motile	+	+	*Bacillus cereus*
TC¹UB	+	Rods	-	+	Motile	+	+	*Bacillus cereus*

Key ; += positive result -= negative result

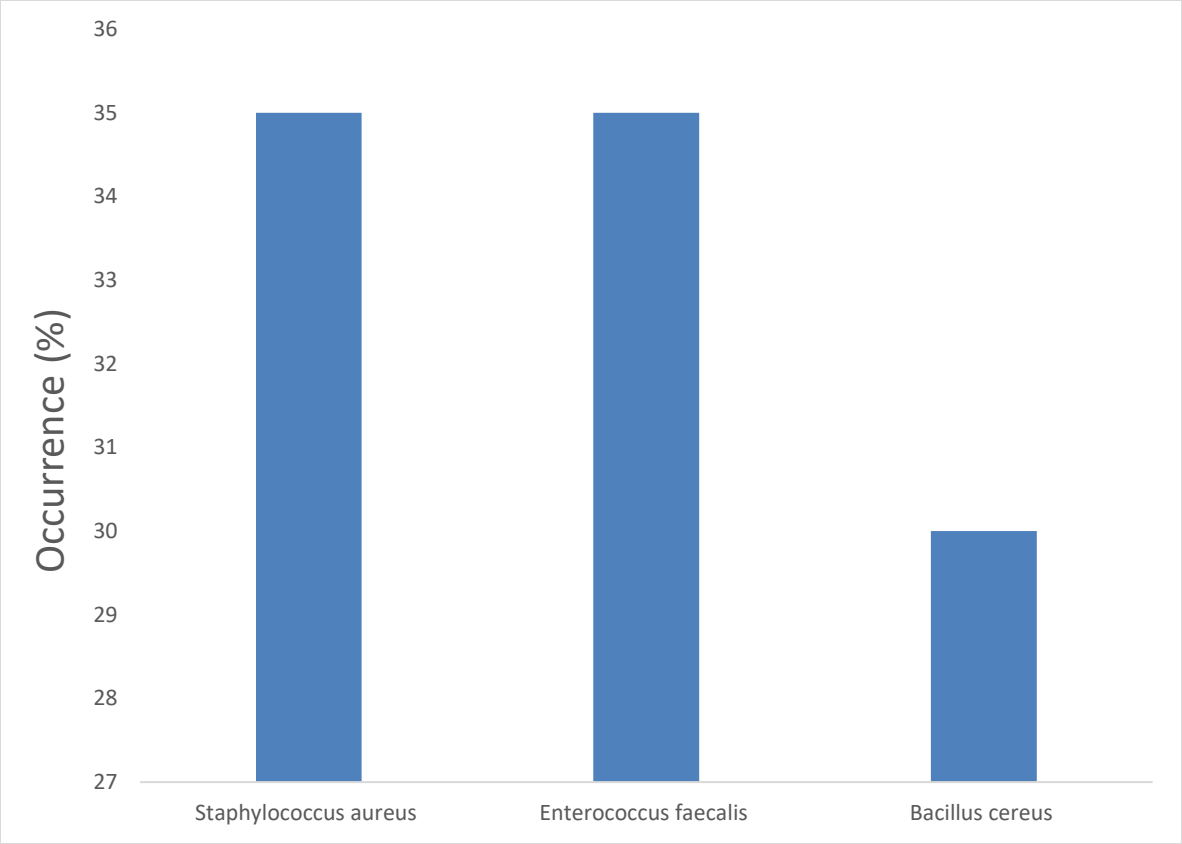

Figure 3.4: Occurrence (%) of Gram-Positive bacteria in tigernut milk

Table 3.3: **Antibiotics resistance of Gram-positive bacteria in tigernut milk**

Bacterial isolates	Number of isolates	ERY (%)	AMX (%)	CPX (%)	OFL (%)	AZY (%)	PEF (%)	LEV (%)	TEX (%)	COT (%)	CEF (%)
Staphylococcus aureus	7										
		6(85)	6(85)	0(0)	1(14)	0(0)	0(0)	0(0)	1(14)	5(71)	7(100)
R		1(14)	0(0)	7(100)	6(85)	3(42)	6(85)	7(100)	5(71)	1(14)	0(0)
I		0(0)	1(14)	0(0)	0(0)	4(57)	1(14)	0(0)	1(14)	1(14)	0(0)
S											
Enterococcus faecalis	7										
		5(71)	6(85)	0(0)	0(0)	0(0)	1(14)	0(0)	1(14)	5(71)	7(100)
R		2(28)	0(0)	6(85)	7(100)	3(42)	6(85)	7(100)	5(71)	0(0)	0(0)
I		0(0)	1(14)	1(14)	0(0)	4(57)	0(0)	0(0)	1(14)	1(14)	0(0)
S											
Bacillus cereus	6										
		5(83)	3(50)	1(16)	1(16)	0(0)	0(0)	0(0)	0(0)	4(66)	5(83)
R		1(16)	0(0)	5(83)	4(66)	1(16)	4(66)	6(100)	2(33)	0(0)	1(16)
I		0(0)	3(50)	0(0)	1(16)	5(83)	2(33)	0(0)	4(66)	2(33)	0(0)
S											

Key: S= Susceptibile , R = resistance, I = intermediate

ERY=Erythromycin,,TEX=Tetracycline,

LEV=Levofloxacin,PEF=Perfloxacin,AMX=Amoxicillin,COT=Cotrimazole,AZY=Aztthromycin,OFL=Ofloxacin,

CEF=Ceftriazone,CPX=Ciprofloxacin

CLSI interpretative standard for antimicrobial disk susceptibility testing:

Antiibiotics	Conc(µg)	S	I	R
ERY	15	≥18	14-17	≤13
AMX	15	≥17	-	≤16
CPX	5	≥21	16-20	≤15
OFL	5	≥18	15-17	≤14
AZY	15	≥ 18	14-17	≤13
PEF	5	≥21	16-20	≤15
LEV	5	≥19	16-18	≤ 15
TEX	30	≥19	15-18	≤14
COT	25	≥16	11-15	≤10
CEF	30	≥25	20-24	≤19

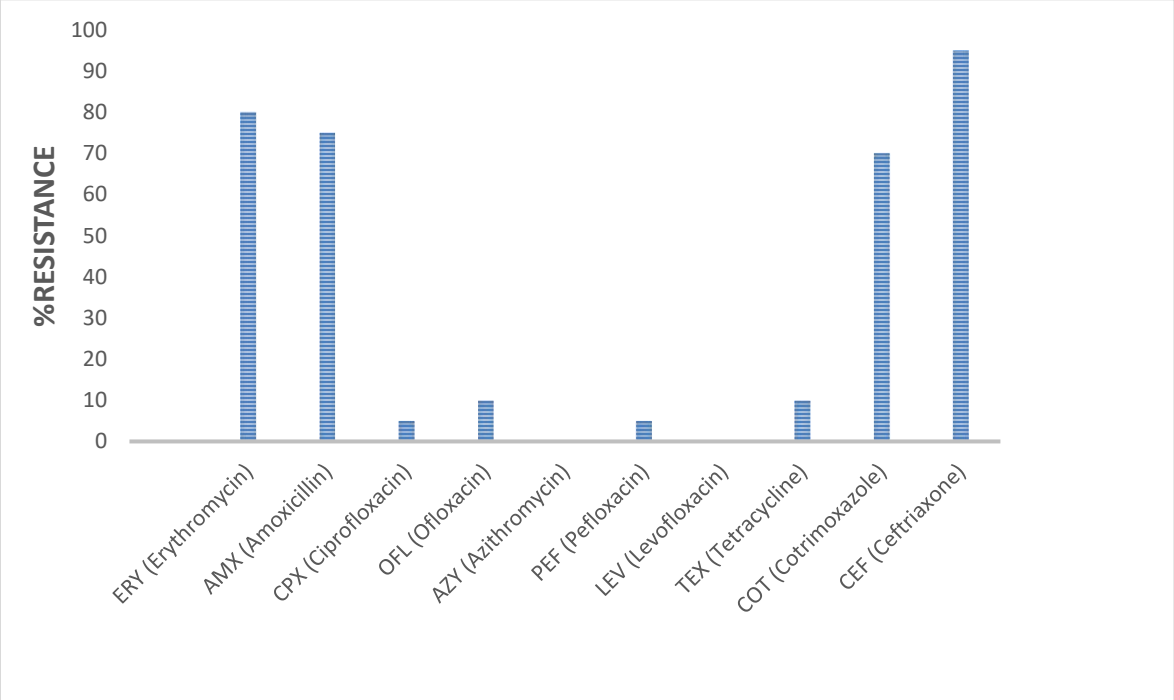

Figure 3.5: Antibiotic resistance pattern of bacterial isolates

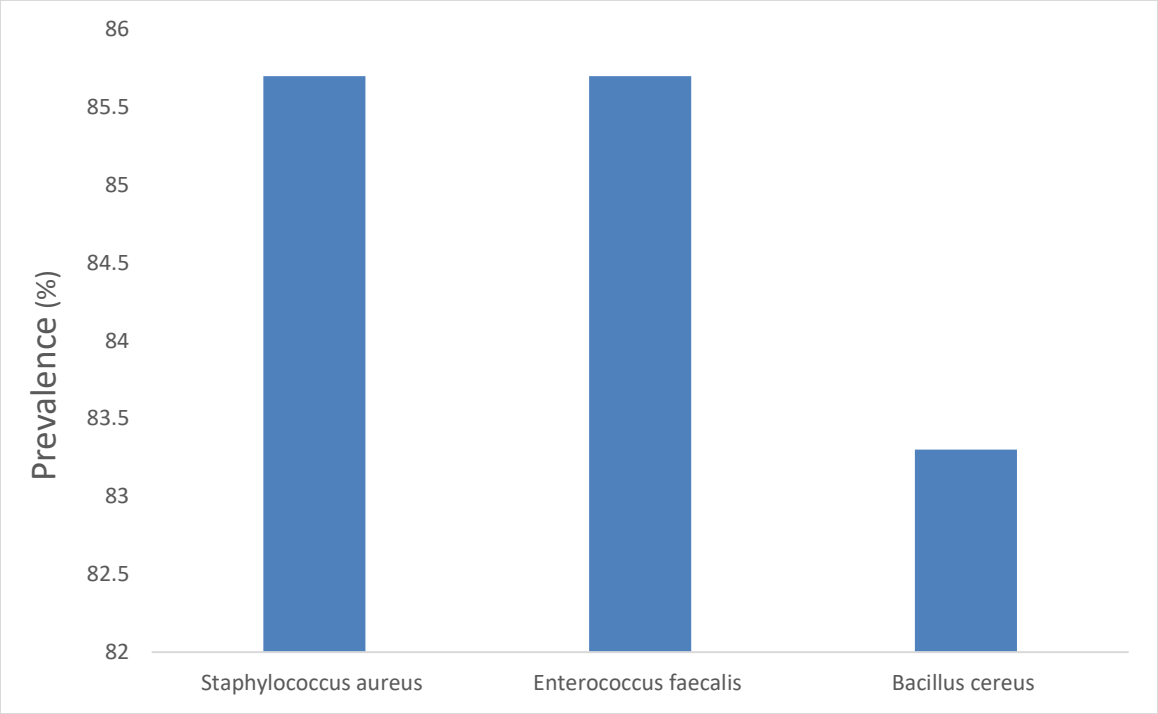

Figure 3.6: Multidrug resistance pattern of Gram-Positive bacteria isolates from tigernut milk

CHAPTER FOUR

DISCUSSION

The comparative analysis of tigernut milk samples, both branded and unbranded, evaluated across Mannitol Salt Agar (MSA), Bile Esculin Agar (BEA), and *Bacillus cereus* agar (BCA) revealed no statistically significant differences in mean colony counts ($p>0.05$ for all). This finding suggests that branding and commercial packaging do not inherently provide an advantage in terms of microbial quality. The observed similarity in bacterial loads between the two categories indicates that contamination may stem from shared factors including raw materials, water quality, processing equipment, and post-production handling (Oyedum and Agbala, 2022).

The systematic collection of samples from various locations in Lagos facilitated the coverage of a diverse array of retail points. This methodological approach enhances the representativeness of the findings and substantiates broader conclusions regarding microbial contamination in tigernut milk distributed within the study area (Pandukur *et al*., 2019). The use of selective and differential media yielded distinct colony morphologies for each bacterial group, which were subsequently validated through standard biochemical tests.

Colonies of *S. aureus* on MSA exhibited a golden yellow colouration and tested positive for coagulase; *E. faecalis* on BEA demonstrated esculin hydrolysis; *B. cereus* colonies on BCA were characterized by a milky zone with pink appearance, and motility tests returned positive results. These findings align with established descriptions in standard microbiological methodologies (Cheesbrough, 2010), thereby confirming the reliability of the identification process. The isolation rates of *S. aureus* (35%), *E. faecalis* (35%), and *B. cereus* (30%) from tigernut milk suggest potential contamination from human handlers, environmental sources, and processing equipment. *S. aureus* is frequently associated with inadequate personal hygiene among food handlers and can precipitate food poisoning via heat-stable enterotoxins (Kadariya *et al*., 2014). *E. faecalis*, a normal gut commensal, may infiltrate food supplies through contamination of water or equipment, while *B. cereus* is known to form heat-resistant spores that can withstand pasteurization and generate toxins (Ramarao *et al*., 2020). The prevalence noted in this investigation is comparable to previous findings concerning street-vended beverages in Nigeria

(Oyedum and Agbala, 2022). Multidrug resistance was identified in 85% of isolates, with *S. aureus* and *E. faecalis* each exhibiting a MDR prevalence of 85.7%, and *B. cereus* showing 83.3%. This elevated rate poses a substantial public health challenge, as MDR bacteria present in ready-to-drink beverages may act as vehicles for the transmission of resistance genes (Capita and Alonso-Calleja, 2013).

The widespread occurrence of resistance is potentially attributable to the indiscriminate use of antibiotics in both human and veterinary contexts, which contributes to environmental contamination and the exertion of selection pressure (Aminov, 2010). Resistance rates recorded against ceftriaxone (95%), erythromycin (80%), and amoxicillin (75%) were particularly pronounced, indicating a diminished efficacy of commonly prescribed antibiotics. This observation is consistent with findings from other studies conducted in Nigeria which documented high resistance rates of foodborne Gram-positive bacteria to β-lactams and macrolides (Oyedum and Agbala, 2022). Conversely, lower resistance rates against ciprofloxacin (5%), ofloxacin (10%), perfloxacin (5%) and levofloxacin (0%) suggest that fluoroquinolones continue to maintain effectiveness against numerous isolates; however, the emergence of resistance to this class of antibiotics underscores the necessary caution in their application.

4.1 CONCLUSION AND RECOMMENDATION

The investigation elucidated the existence of antibiotic-resistant Gram-positive bacteria, (*Staphylococcus, Enterococcus*, and *Bacillus cereus*,)in tigernut milk. This finding accentuates the potential health hazards linked to the consumption of contaminated tigernut milk, particularly in instances where hygienic practices during production and handling are suboptimal. The results emphasize the critical need for the surveillance of microbial quality in beverages consumed locally to protect public health. Producers of tigernut milk are urged to implement rigorous hygienic protocols during the stages of processing, packaging, and storage. Regulatory food safety authorities must mandate regular microbiological assessments of tigernut milk available in stores and markets. Ongoing surveillance of antibiotic resistance patterns in foodborne bacteria should be promoted to inform treatment strategies and safeguard public health. Comprehensive investigations should be conducted on antimicrobial resistance in street-vended beverages to formulate improved control measures.

REFERENCES

Abraha, A., Bikila, T., Abemu, S., and Muktar, Y. (2017). *Bacillus cereus* Isolation and load from cow milk sold in Market of Haramaya District, eastern Ethiopia. *International Journal of Food Contamination* **4:** 1- 6.

Adejuyitan, J. A. (2011). Tigernut processing: its food uses and health benefits. *American Journal of Food Technology* **6**(3)**:** 197-201.

Adewunmi, O. O., Adetunji, M. C., Fagbemi, O. T., Aransiola, M. N., Odunlade, A. K., Fowora, M. A., and Omoyajowo, A. O. (2024). A Survey of Fungal Contamination and Their Metabolites in Street-Vended Tiger Nuts and Dates in Lagos State, Nigeria Using LCMS/MS Analysis. *Covenant Journal of Health and Life Sciences* **2**(2)**:** 1-12

Afusat, S.A.(2022). Isolation and Identification of bacteria in tigernut *(cyperus esculentus*) drink from selected areas in Lagos state.

Al-joda, B.M.S., and Jasu, A.H. (2021). Biochemical testing revision for identification several kinds of bacteria. *Journal of University of Babylon for Pure and Applied sciences* **29** (2)**:**168-176.

Aminov, R.I. (2010). A brief history of the antibiotic era: lessons learned and challenges for the future. *Frontier in microbiology* **1:**134.

Awadelkareem, A.B. (2006). Study On Enterococci And *Bacillus cereus* isolated from minced meat (Doctoral dissertation, University of Khartoum).

Belewu, M. A., and Abodunrin, O. A. (2008). Preparation of Kunnu from unexploited rich food source: Tiger nut (*Cyperus esculentus*). *Pakistan Journal of Nutrition* **7**(1)**:** 109-111.

Capita, R., and Alonso-Calleja,C .(2013). Antibiotic-resistant bacteria: a challenge for the food industry. *Critical reviews in food science and nutrition* **53**(1)**:** 11- 48.

Cheesbrough, M. (2010). Microbiological techniques. *In District laboratory practice in tropical countries* (Update, Part 1). Cambridge University Press.

Coskuner, Y., Ercan, R., Karababa, E., and Nazlican, A. N. (2002). Physical and chemical properties of chufa (*Cyperus esculentus L*) tubers grown in the Cukurova region of Turkey. *Journal of the Science of Food and Agriculture* **82**(6)**:** 625 - 631

Fisher, K., and Phillips, C. (2009). The ecology, epidemiology and virulence of *Enterococcus*. *Microbiolog*y **155**(6)**:** 1749-1757.

Gambo,A ., and Da' u, A. (2014). Tigernut (*cyperus esculentus*): Composition, products, uses and health benefits - a review.*Bayero Journal of pure and applied Sciences* **7**(1)**:**56 -61.

Hayatu, K. B. (2022). A Project submitted in Partial Fulfillment of the Requirements.

Hollenbeck, B.L., and Rice, L.B. (2012).Intrinsic and acquired resistance mechanisms in *Enterococcus*. *Virulence* **3**(5)**:** 421-569.

Iyiola, C.O. (2022). Bacteriological Quality Of Kunu Drinks Sold in Alimosho Local Government Area, Lagos State, Nigeria (Doctoral Dissertation, Caleb University).

Kadariya, J., Smith, T. C., and Thapaliya, D. (2014). *Staphylococcus aureus* and staphylococcal food-borne disease: an ongoing challenge in public health. *BioMed research international* **2014**(1)**:**827965.

Klinker , K . P., Hidayat, L.K., DeRyke, C.A., Depestoh, D.D., Moty, R., and Bauer, K.A. (2021). Antimicrobial stewardship and antibiograms: importance of moving beyond traditional antibiograms. *Therapeutic Advances in Infectious Diseases* **8:**204993612110211373

Maduka, N., Etamishere, E. E., and Ahaotu, I. (2022). Assessment of Microbiological and Nutritional Quality of Some Local Brands of Tigernut-Milk Sold in Benin City, Nigeria. *Nigerian annals of pure and applied sciences* **5(**2)**:** 381-397.

Magiorakos, A. P., Srinivasan, A., Carey, R. B., Carmeli, Y., Falagas, M. E., Giske, C. G., and Monnet, D. L. (2012). Multidrug-resistant, extensively drug-resistant and pandrug-resistant bacteria: an international expert proposal for interim standard definitions for acquired resistance. *Clinical microbiology and infection* **18**(3)**:** 268-281.

Majumdar, A., Pradhan, N., Sadasivan, J., Acharya, A., Ojha, N., Babu, S., and Bose, S. (2018). Food degradation and foodborne diseases: A microbial approach. *In Microbial contamination and food degradation* (pp. 109-148). Academic Press.

Makanjuola, D.O., Zamani, P.A., Adegbite, I. A., Oluyomi, C.D., and Ahmadu, J.H.(2022). Bateriological Properties of Tigernut Drink (kunu-aya) sold in Shehu Idris College of Health Science and Technology, Makarfi , Kaduna State, Nigeria. *African Scholar Journal of Biotechnology and Agricultural Research* **26**(1)**:** 145-152.

Mbachu, A. E., Etok, C. A., Agu, K. C., Okafor, O. I., Awah, N. S., Chidi-Onuorah, L. C., ... and Ikele, M. O. (2014). Microbial quality of kunu drink sold in Calabar, Cross River State, Nigeria. *Journal of Global Biosciences* **3**(2)**:**511-515.

Motlhanka , K., Zhou, N., and Lebami, K. (2018). Microbial and chemical diversity of traditional non-cereal based alcoholic beverages of sub- Saharan Africa. *Beverages* **4**(2)**:** 36.

N odu, N., O Amadi, L., and O Aguanah, E. (2021). Screening of bacteria isolates with probiotic potential of tiger nut drink, ogi and palmwine. *Journal of Applied Life Science International* **24**(7)**:** 36-48

Ndukwe, J. k., A duba , C.C., Ughamba , K.T., Chukwu, K.O., Eze , C.N., Nwaiwu, O., and Onyeaka, H. (2023). Diet diversification and priming with kunu: An indigenous probiotic cereal-based non-alcoholic beverage in Nigeria. *Beverages* **9**(1)**:** 14.

Nwachukwu, I. O. (2024). Assessment of Microbiological Quality of Commercially Prepared Ready to Drink Tigernuts Sold in Owerri, Imo State. *Int'l Journal of Education Research and Scientific Development* **5**(5)**:** 13-13.

Ocheme, O. B., Ariahu, C. C., and Kongo Ingbian, E. (2013). Moisture sorption characteristics of Dakuwa (Nigerian cereal/groundnut snack). *International journal of food engineering* **9**(4)**:**499-504.

Ogbonna, A. C., Abuajah, C.I., and Utuk, A .R. (2013). Tigernut milk: A nutritious but under utilized food ingredient. *Food Biology*. **2**(2)**:**14-17.

Oladele, A. K., and Aina, J. O. (2007). Chemical composition and functional properties of flour produced from two varieties of tigernut (*Cyperus esculentus*). *African Journal of Biotechnology* **6**(21)**:**2473-2476.

Oladele, A. K., Ibanga, U. I., and Adebesin, O. L. (2009). Effect of substituting maize with tigernut on the quality and acceptability of Dakuwa. *In Proceedings of the 33rd Annual conference and General Meeting of Nigerian Institute of Food Science and Technology, Yola, Nigeria.*

Oladele, A. K., Ntuem, I. G., Liman, A. A., and Adewoyin, T. A. (2013). Effects of roasting temperature of tigernut (*Cyperus esculentus L.*) on acceptability of Dakuwa. *In Proceedings of the 37th Annual conference and General Meeting of Nigerian Institute of Food Science and Technology, Abuja, Nigeria* (pp. 133-134).

Oyedum, U.M., and Agbala, J. (2022). The study of multidrug resistant bacteria from locally produced tigernut drinks (kunu-aya) sold in Minna, Niger state. *Scientia Aficana* **21**(2)**:** 91-96.

Pandukur, S.G., Danshak, D.T., Plangnan, G.A., Bala, D.A., and Helina, J. U. (2019). Antibiogram of some bacteria isolated from processed tiger-nut beverage (kunu-aya) sold in the University of Jos Community Plateau State Nigeria. *Direct Research Journal of Agriculture and Food Science* **7**(11)**:** 313-318.

Petruzzi, L., Corbo, M. R., Sinigaglia, M., and Bevilacqua, A. (2017). Microbial spoilage of foods: Fundamentals. In *The microbiological quality of food* (pp. 1-21). Woodhead Publishing.

Ramarao N, Tran S.L., Marin M., Vidic J. (2020). Advanced Methods for Detection of *Bacillus cereus* and Its Pathogenic Factors **20**(9)**:** 2667.

Reller , L.B., Weinstein, M., Jorgesen, J.H., and Feraro, M .J .(2009) Antimicrobial susceptibility testing: a review of general principles and contemporary practices. *Clinical infectious diseases* **49**(11)**:** 1749-1755.

Sànchez-Zapata, E., Fernandez-Lopez ,J., and Angel Perez- Alvarez , J. (2022). Tiger nut (*cyperus esculentus*) commercialization: health aspects , composition, properties, and food application. *Comprehensive Reviews in Food science and Food safety* **11**(4)**:** 366-377.

Sanful, R. E. (2009). The use of tiger-nut (*Cyperus esculentus*) cow milk and their composite as substrates for yoghurt production. *Pakistan Journal of Nutrition* **8**(6)**:** 755-758.

Sherman, J. M. (1937). The Streptococci. *Bacteriological reviews* **1**(1)**:** 3-97.

APPENDIX I

Table 1 : Colony count for unbranded tigernut milk on mannitol salt agar

Sample	Colony count (CFU/mL)
TM^1UA	TNC
TM^2UA	1.025×10^8
TM^3UA	TNC
TM^4UA	8×10^7
TM^5UA	TNC
TM^1UB	3.2×10^7
TM^2UB	5.2×10^7
TM^3UB	4.8×10^7
TM^4UB	5×10^7
TM^5UB	6.8×10^7

Table 2: Colony count for branded tigernut milk on mannitol salt agar

Sample	Colony count (CFU/mL)
TM^1BA	8×10^6
TM^2BA	1.225×10^8
TM^3BA	4×10^6
TM^4BA	1.175×10^8
TM^5BA	9.6×10^7
TM^1BB	2.7×10^7
TM^2BB	3.1×10^7
TM^3BB	4.45×10^7

TM^4BB	9.3×10^7
TM^5BB	No growth
TM^1BC	8.4×10^7
TM^2BC	6.6×10^7
TM^3BC	5.8×10^7
TM^4BC	7.8×10^7
TM^5BC	9 x107

Table 3: Colony count for unbranded tigernut milk on bile esculin agar

Sample	Colony count (CFU/mL)
TE^1UA	TNC
TE^2UA	8.5×10^7
TE^3UA	1.3×10^8
TE^4UA	1.5×10^8
TE^5UA	1.15×10^8
TE^1UB	4.4×10^7
TE^2UB	7.4×10^7
TE^3UB	6.2×10^7
TE^4UB	7.2×10^7
TE^5UB	9.2×10^7

Table 4: Colony count for branded tigernut milk on bile esculin agar

Sample	Colony count (CFU/mL)
$TE^{1}BA$	8.15×10^{7}
$TE^{2}BA$	No growth
$TE^{3}BA$	9.9×10^{7}
$TE^{4}BA$	5.1×10^{7}
$TE^{5}BA$	No growth
$TE^{1}BB$	3.75×10^{7}
$TE^{2}BB$	1.05×10^{7}
$TE^{3}BB$	9.15×10^{7}
$TE^{4}BB$	1.15×10^{7}
$TE^{5}BB$	1.42×10^{7}
$TE^{1}BC$	1.4×10^{8}
$TE^{2}BC$	1.04×10^{8}
$TE^{3}BC$	9.4×10^{7}
$TE^{4}BC$	1.12×10^{8}
$TE^{5}BC$	1.28×10^{8}

Table 5: Colony count for unbranded tigernut milk on *Bacillus cereus* agar

Sample	Colony count (CFU/mL)
$TC^{1}UA$	TNC
$TC^{2}UA$	TNC
$TC^{3}UA$	TNC
$TC^{4}UA$	5×10^{7}

TC^5UA	TNC
TC^1UB	$7x\ 10^7$
TC^2UB	$8.4\ x\ 10^7$
TC^3UB	$4.5\ x\ 10^7$
TC^4UB	$8.8\ x\ 10^7$
TC^5UB	$1.16\ x\ 10^8$

Table 6: Colony count for branded tigernut milk on *Bacillus cereus* agar

Sample	Colony count (CFU/mL)
TC^1BA	$6.35\ x\ 10^7$
TC^2BA	$2.2\ x\ 10^8$
TC^3BA	$1\ X\ 10^7$
TC^4BA	$1.2\ X\ 10^8$
TC^5BA	No growth
TC^1BB	No growth
TC^2BB	$6\ x\ 10^7$
TC^3BB	$3.9\ x\ 10^7$
TC^4BB	$1.10\ x\ 10^8$
TC^5BB	$9.9\ x\ 10^7$
TC^1BC	$1.02\ x\ 10^8$
TC^2BC	$1.42\ x\ 10^8$
TC^3BC	$8\ x\ 10^7$
TC^4BC	$1.35\ x\ 10^8$
TC^5BC	$1.52\ x\ 10^8$

Key: UA=Unilag ,UB= Agege, BA= Bariga, BB= Bariga, BC= Abulegba

APPENDIX II

STATISTICAL ANALYSIS OF BACTERIAL LOAD ON THE DIFFERENT MEDIA

MANITTOL SALT AGAR (MSA)

Table IIA: Statistical analysis of bacterial load on manittol salt agar

Group Statistics

Brand Type		N	Mean	Std. Deviation	Std. Error Mean
Colony Count	Branded	14	65750000.0000	38335939.15335	10245710.70800
	Unbranded	7	61857142.8571	23737653.43239	8971989.67005

Independent Samples Test

		Levene's Test for Equality of Variances		t-test for Equality of Means						
		F	Sig.	t	df	Sig. (2-tailed)	Mean Difference	Std. Error Difference	95% Confidence Interval of the Difference Lower	Upper
Colony Count	Equal variances assumed	2.523	0.129	0.244	19	0.810	3892857.14286	15924952.80721	-29438452.14795	37224166.43367
	Equal variances not assumed			0.286	17.846	0.778	3892857.14286	13618780.65583	-24736889.03220	32522603.31791

Independent Samples Effect Sizes					
		Standardizer[a]	Point Estimate	95% Confidence Interval	
				Lower	Upper
Colony Count	Cohen's d	34401829.92597	0.113	-0.796	1.020
	Hedges' correction	35838586.19836	0.109	-0.764	0.979
	Glass's delta	23737653.43239	0.164	-0.754	1.069

The denominator used in estimating the effect sizes.

Cohen's d uses the pooled standard deviation.

Hedges' correction uses the pooled standard deviation, plus a correction factor.

Glass's delta uses the sample standard deviation of the control group.

BILE ESCULIN AGAR (BEA)

Table IIB: Statistical analysis of bacterial load on bile esculin agar

Group Statistics					
Brand Type		N	Mean	Std. Deviation	Std. Error Mean
Colony Count	Branded	13	74976923.0769	45061886.22044	12497918.56438
	Unbranded	9	91555555.5556	34117851.30658	11372617.10219

Independent Samples Test										
		Levene's Test for Equality of Variances		t-test for Equality of Means						
		F	Sig.	T	Df	Sig. (2-tailed)	Mean Difference	Std. Error Difference	95% Confidence Interval of the Difference	
									Lower	Upper
Colony Count	Equal variances assumed	1.819	0.193	-0.932	20	0.363	-16578632.47863	17794413.75635	-53697129.13990	20539864.18263

	Equal variances not assumed	-0.981	19.769	0.338	-16578632.47863	16897762.81627	-51853176.91365	18695911.95639

Independent Samples Effect Sizes

		Standardizer[a]	Point Estimate	95% Confidence Interval	
				Lower	Upper
Colony_Count	Cohen's d	41036023.99060	-0.404	-1.258	0.460
	Hedges' correction	42659449.22107	-0.389	-1.210	0.442
	Glass's delta	34117851.30658	-0.486	-1.354	0.409

The denominator used in estimating the effect sizes. Cohen's d uses the pooled standard deviation. Hedges' correction uses the pooled standard deviation, plus a correction factor. Glass's delta uses the sample standard deviation of the control group.

***Bacillus cereus* AGAR (BCA)**

Table IIC: Statistical analysis of bacterial load on *Bacillus cereus* agar

Group Statistics

Brand Type		N	Mean	Std. Deviation	Std. Error Mean
Colony Count	Branded	13	102500000.0000	54520638.29414	15121304.38004
	Unbranded	6	75500000.0000	26379916.60336	10769555.85590

Independent Samples Test

		Levene's Test for Equality of Variances		t-test for Equality of Means						
		F	Sig.	T	df	Sig. (2-tailed)	Mean Difference	Std. Error Difference	95% Confidence Interval of the Difference Lower	Upper
Colony Count	Equal variances assumed	1.860	0.190	1.140	17	0.270	27000000.00000	23684714.19026	-22970378.95515	76970378.95515
	Equal variances not assumed			1.454	16.854	0.164	27000000.00000	18564406.25194	-12193349.54794	66193349.54794

Independent Samples Effect Sizes

		Standardizer[a]	Point Estimate	95% Confidence Interval	
				Lower	Upper
Colony Count	Cohen's d	47988662.87683	0.563	-0.431	1.540
	Hedges' correction	50244042.28698	0.537	-0.411	1.471
	Glass's delta	26379916.60336	1.024	-0.158	2.136

The denominator used in estimating the effect sizes. Cohen's d uses the pooled standard deviation. Hedges' correction uses the pooled standard deviation, plus a correction factor. Glass's delta uses the sample standard deviation of the control group.

Table IID :Comparison of bacterial load in branded and unbranded tigernut milk

Media	**Probable microrganisms**	**Mean of colony count of branded tigernut milk (CFU mL)**	**Mean of colony count of unbranded tigernut milk (CFU /mL)**	**Significant Difference**	**Microbiologically safe**
MSA	*Staphylococcus* spp.	65.8	61.9	P = 0.810	Similar
BEA	*Enterococcus* spp.	74.9	91.6	P = 0.363	Branded
BCA	*Bacillus cereus*	102.5	75.5	P = 0.270	Unbranded

Key: MSA -Mannitol Salt Agar, BEA - Bile Esculin Agar, BCA - *Bacillus cereus* Agar

[a] Mannitol Salt Agar (MSA) – *Staphylococcus aureus*

Branded tigernut milk had mean colony count of 65.8 CFU/mL and unbranded tigernut milk had a mean colony count of 61.9 CFU/mL. There is no statistically significant difference observed between the branded and unbranded tigernut milk samples (p=0810).

Both branded and unbranded tigernut milk exhibit equivalent levels of contamination with *Staphylococcus aureus.* Given that *S. aureus* possesses pathogenic potential, this finding suggests a potential food safety concern if toxin-producing strains are present; however, both categories display analogous contamination levels.

[b] Bacillus Cereus Agar (BCA) – *Bacillus cereus*

Branded tigernut milk had mean colony count of 102.5 CFU /mL and unbranded tigernut milk had 75.5 CFU/mL

There is no statistically significant difference observed between branded and unbranded tigernut milk samples (p=0.270).

Bacillus cereus was found to be more prevalent in branded tigernut milk relative to unbranded tigernut milk. Considering that *B. cereus* is a spore-forming bacterium associated with foodborne illness, this implies that the branded milk may present a heightened risk for toxin-related foodborne diseases. The table designates unbranded samples as “microbiologically safe” in this context; however, it is important to recognize that both types could pose safety risks if microbial counts surpass acceptable level.

[c] Bile Esculin Agar (BEA) – *Enterococcus faecalis*

Branded tigernut milk had mean colony count of 74.9 CFU/mL and unbranded tigernut milk had 91.6 CFU/mL.

There is no statistically significant difference observed between branded and unbranded tigernut milk samples (p=0.363).

Both branded and unbranded samples were found to contain *Enterococcus faecalis*. Although not statistically distinct, unbranded tigernut milk exhibited higher microbial counts. Enterococci serve as indicators of fecal contamination; thus, this finding implies potential hygiene deficiencies during the production process. The table categorizes branded samples as "microbiologically safe," presumably due to their comparatively lower mean counts.

Both branded and unbranded tigernut milk were found to harbor potential foodborne pathogens (*S. aureus, E. faecalis, B. cereus*).Statistical analysis revealed no significant differences in

contamination levels between branded and unbranded tigernut milk ($p > 0.05$ across all instances).

The designation of "microbiologically safe" in the table appears to be relative—indicating which sample type presented lower mean counts—but it is critical to note that neither type can be regarded as genuinely safe if pathogens are detected.

Appendix III

STATISTICAL ANALYSIS ON ANTIBIOTIC SUSCEPTIBILITY TEST OF BACTERIAL ISOLATES

Table IIIA: Statistical analysis of Erythromycin on bacterial isolates

Erythromycin					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Susceptible (S)	1	5.0	5.0	5.0
	Resistant (R)	16	80.0	80.0	85.0
	Intermediate (I)	3	15.0	15.0	100.0
	Total	20	100.0	100.0	

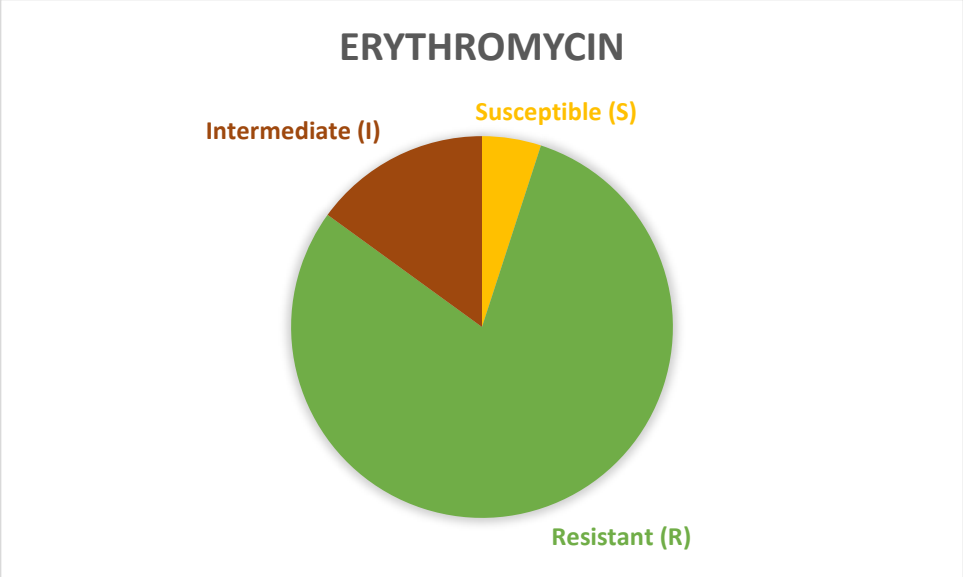

Table IIIB : Statistical analysis of Amoxicillin on bacterial isolates

Amoxicillin					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Susceptible (S)	4	20.0	20.0	20.0
	Resistant (R)	16	80.0	80.0	100.0
	Total	20	100.0	100.0	

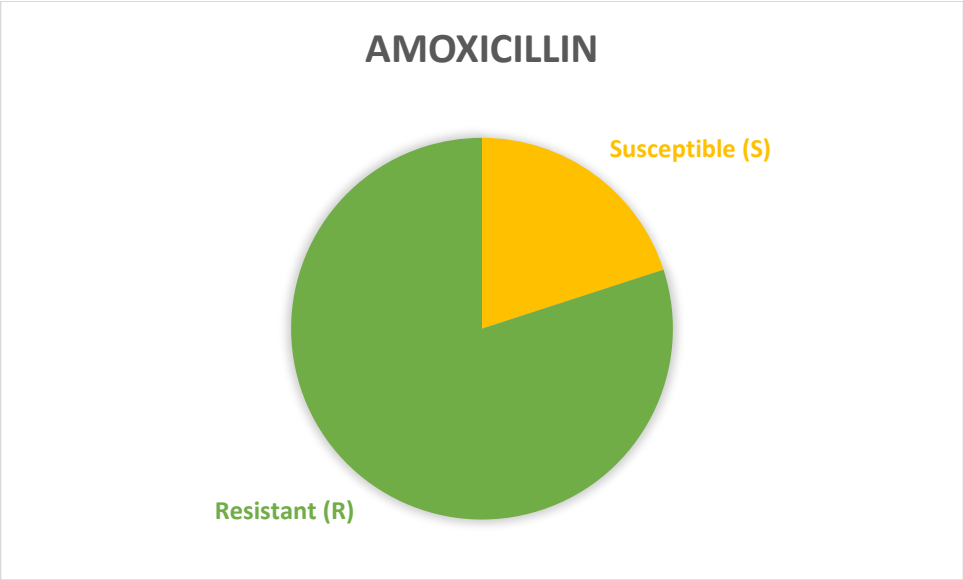

Table IIIC: Statistical analysis of Ciprofloxacin on bacterial isolates

Ciprofloxacin					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Susceptible (S)	1	5.0	5.0	5.0
	Resistant (R)	1	5.0	5.0	10.0
	Intermediate (I)	18	90.0	90.0	100.0
	Total	20	100.0	100.0	

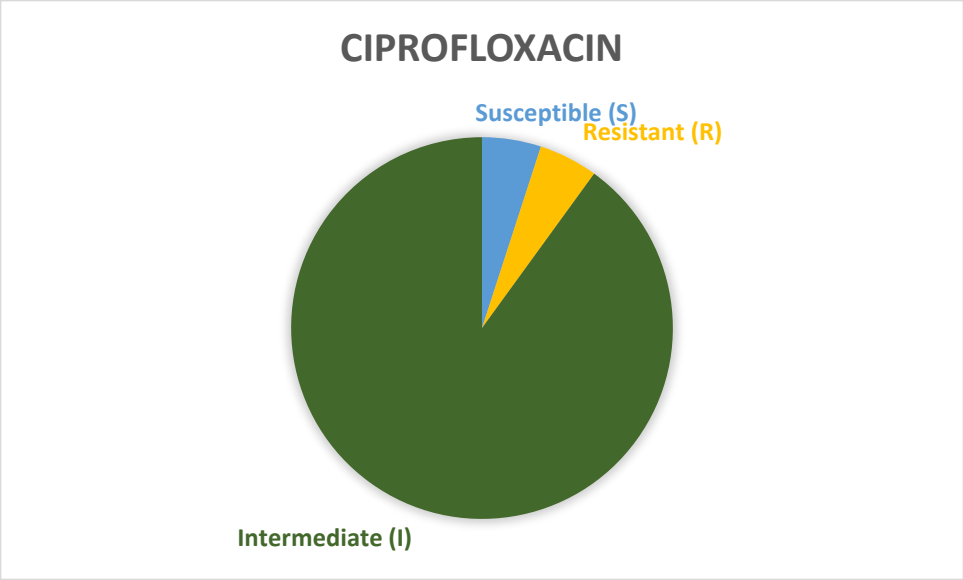

Table IIID: Statistical analysis of Ofloxacin on bacterial isolates

Ofloxacin					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Susceptible (S)	1	5.0	5.0	5.0
	Resistant (R)	2	10.0	10.0	15.0
	Intermediate (I)	17	85.0	85.0	100.0
	Total	20	100.0	100.0	

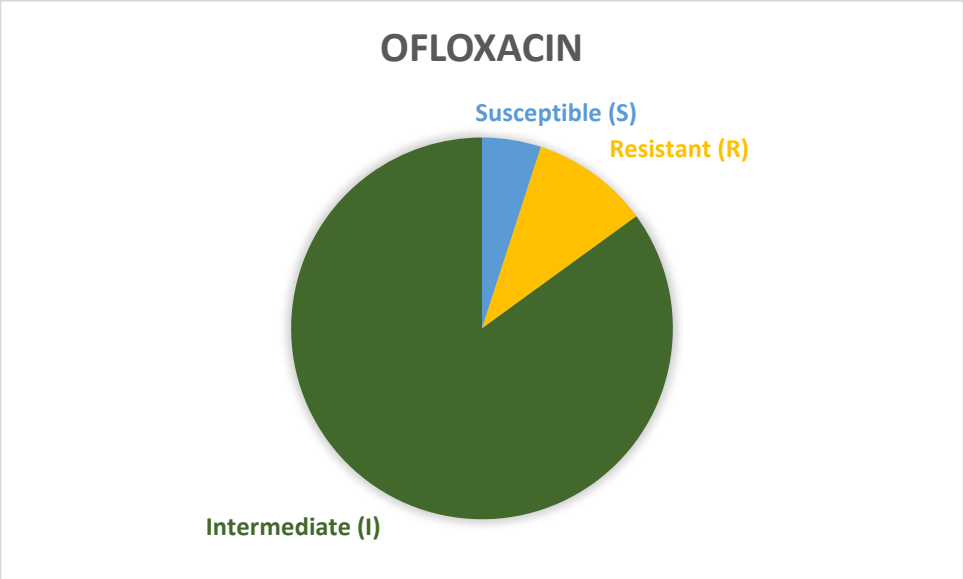

Table IIIE: Statistical analysis of Azithromycin on bacterial isolates

Azithromycin					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Susceptible (S)	13	65.0	65.0	65.0
	Intermediate (I)	7	35.0	35.0	100.0
	Total	20	100.0	100.0	

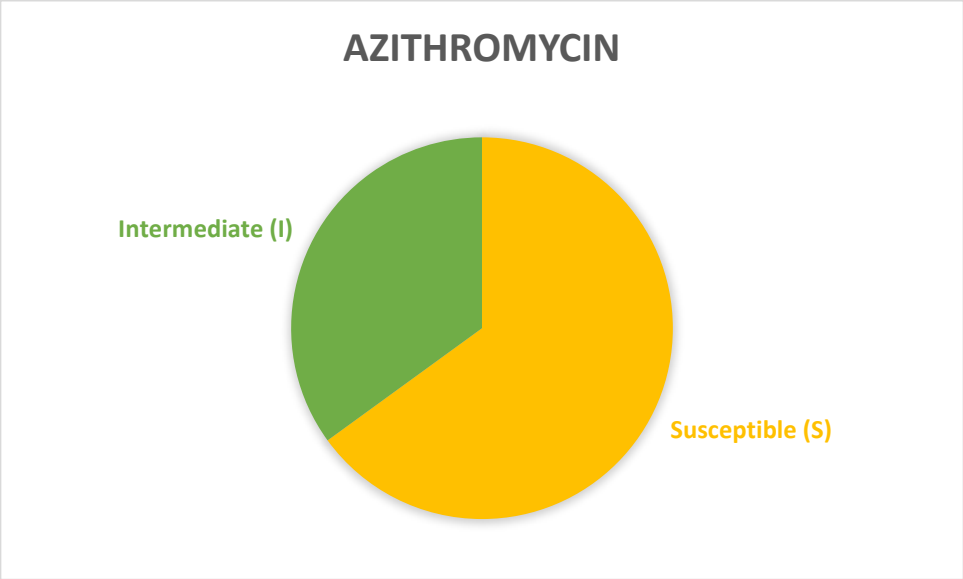

Table IIIF : Statistical analysis of Perfloxacin on bacterial isolates

Perfloxacin					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Susceptible (S)	3	15.0	15.0	15.0
	Resistant (R)	1	5.0	5.0	20.0
	Intermediate (I)	16	80.0	80.0	100.0
	Total	20	100.0	100.0	

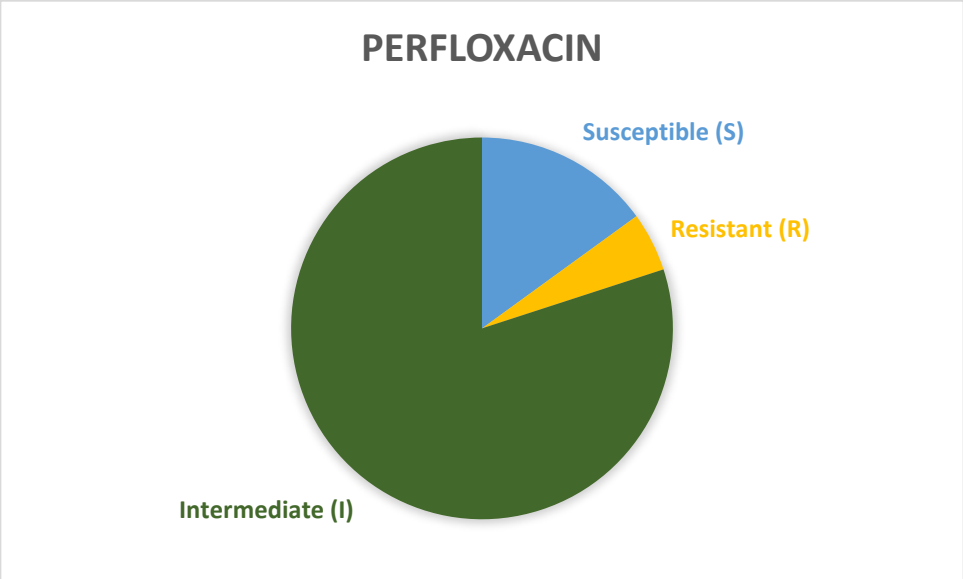

Table IIIG: Statistical analysis of Levofloxacin on bacterial isolates

Levofloxacin					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Susceptible (S)	5	25.0	25.0	25.0
	Intermediate (I)	15	75.0	75.0	100.0
	Total	20	100.0	100.0	

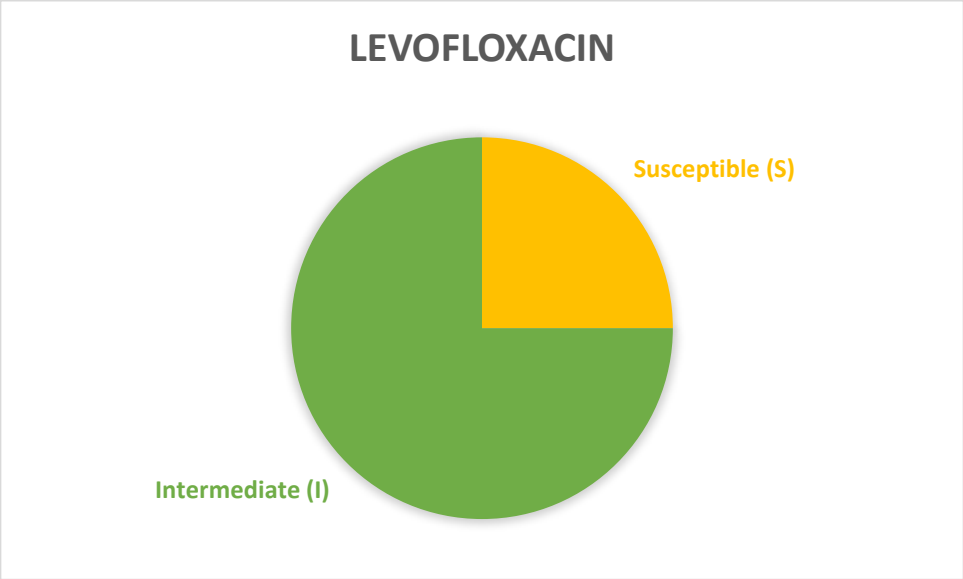

Table IIIH: Statistical analysis of Tetracycline on bacterial isolates

Tetracycline					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Susceptible (S)	6	30.0	30.0	30.0
	Resistant (R)	2	10.0	10.0	40.0
	Intermediate (I)	12	60.0	60.0	100.0
	Total	20	100.0	100.0	

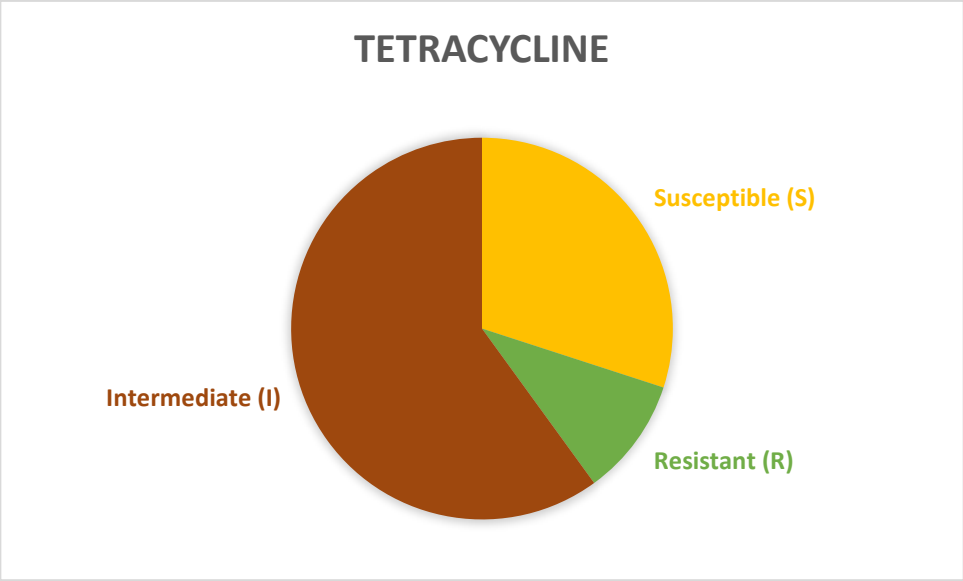

Table IIII: Statistical analysis of Cotrimazole on bacterial isolates

Cotrimazole					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Susceptible (S)	5	25.0	25.0	25.0
	Resistant (R)	14	70.0	70.0	95.0
	Intermediate (I)	1	5.0	5.0	100.0
	Total	20	100.0	100.0	

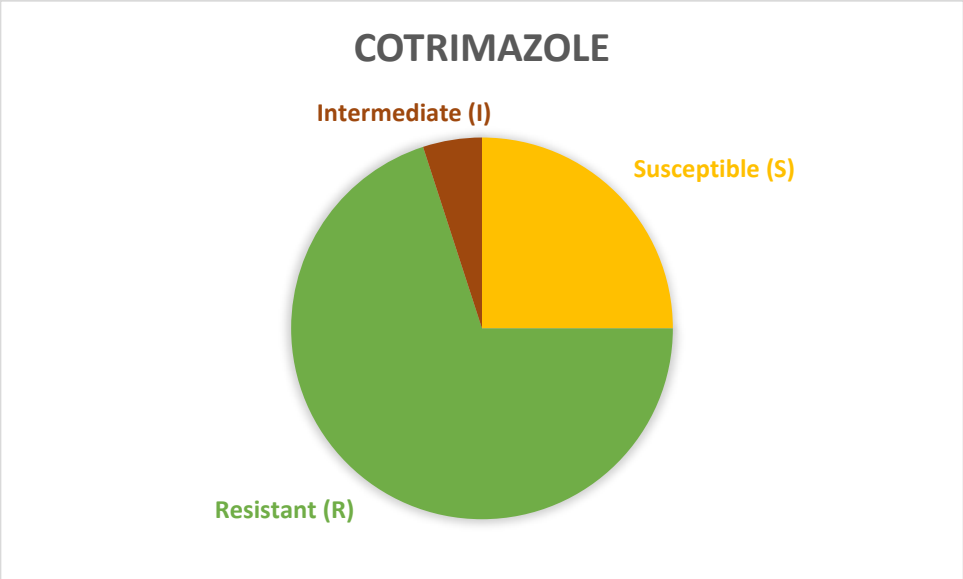

Table IIIJ: Statistical analysis of Ceftriazone on bacterial isolates

CEF					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Resistant (R)	19	95.0	95.0	95.0
	Intermediate (I)	1	5.0	5.0	100.0
	Total	20	100.0	100.0	

[a] Antibiotic Susceptibility Test of Bacterial Isolates

The antibiotic susceptibility profile reveals that a large proportion of the bacterial isolates exhibited high levels of resistance to multiple antibiotics.

Highest resistance rates were observed with Ceftriaxone (CEF) at 95%, Erythromycin (ERY) at 80%, and Amoxicillin (AMX) at 75%. These results suggest that these antibiotics are largely ineffective against the Gram-positive bacteria present in the tigernut milk samples.

Moderate to high resistance was also recorded for Cotrimoxazole (COT) at 70%, and Tetracycline (TEX) at 10% resistance, indicating reduced therapeutic potential.

Ciprofloxacin (CPX),Perfloxacin and Ofloxacin (OFL) showed low resistance (5–10%), but the majority of isolates exhibited intermediate susceptibility (≥85%), suggesting reduced potency and possible emerging resistance.

Azithromycin (AZY) and Levofloxacin (LEV) demonstrated comparatively better activity, with low resistance levels (0%).

The multidrug resistance patterns indicates that many isolates are resistant to at least three antibiotic classes, pointing to the likelihood of antimicrobial misuse in the environment or food chain from which these bacteria originated.

Table IIIK : Frequency of antibiotics resistance pattern of bacterial isolates

Antibiotic	Resistant	% Resistant
ERY (Erythromycin)	16	80%
AMX (Amoxicillin)	15	75%
CPX (Ciprofloxacin)	1	5%
OFL (Ofloxacin)	2	10%
AZY (Azithromycin)	0	0%
PEF (Pefloxacin)	1	5%
LEV (Levofloxacin)	0	0%
TEX (Tetracycline)	2	10%
COT (Cotrimoxazole)	14	70%
CEF (Ceftriaxone)	19	95%

[a] Out of the 20 isolates tested, the highest resistance was recorded against Cef (95%), followed by Erythromycin (80%) and Amoxicillin (75%). Moderate resistance was observed against Cotrimoxazole (70%), while minimal resistance (≤10%) was seen against Ciprofloxacin, Ofloxacin, Perfloxacin and Tetracycline. No resistance was observed against Azithromycin and Levofloxacin.

Total Resistant					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	1	1	5.0	5.0	5.0
	2	2	10.0	10.0	15.0
	3	5	25.0	25.0	40.0
	4	10	50.0	50.0	90.0
	5	2	10.0	10.0	100.0
	Total	20	100.0	100.0	

[a] Out of 20 isolates,17 isolates (85%) are resistant to ≥3 antibiotics, which qualifies as Multidrug Resistant (MDR) under the standard definition (resistance to at least 3 antimicrobial classes).

Only 3 isolates (15%) were resistant to fewer than 3 antibiotics.

The most common resistance level was 4 antibiotics (50% of all isolates), followed by 3 antibiotics (25%).

The distribution suggests a high prevalence of MDR, indicating a serious antibiotic resistance concern.

APPENDIX IV

Table IVA: Occurrence of Gram-Positive bacteria in tigernut milk samples

Bacterial species	Occurrence (Number of positive samples)	Occurrence (%)
Staphylococcus aureus	7	35
Enterococcus faecalis	7	35
Bacillus cereus	6	30
Total	**20**	**100**

APPENDIX V

Table VA: Multidrug resistance pattern of Gram-Positive bacterial isolates from tigernut milk

Bacterial species	Number of isolates	MDR isolates	MDR prevalence (%)
Staphylococcus aureus	7	6	85.7
Enterococcus faecalis	7	6	85.7
Bacillus cereus	6	5	83.3
Total Gram-positive isolates	**20**	**17**	**85**

APPENDIX VI

PICTURES FROM METHODOLOGY

Tigernut milk sample

Enterococcus on Bile Esculin Agar

Staphylococcus aureus on mannitol salt agar

Citrate test

Motility test

Bacillus cereus on *Bacillus cereus* agar

APPENDIX VII

COMPOSITION OF MEDIA

1. **Mannitol Salt Agar**

   Peptone : 10.0g

   Beef Extract : 1.0g

   D-mannitol: 10.0g

   Sodium chloride (Nacl):75.0g

   Agar: 15.0g

   Phenol Red: 0.025g

   Distilled water: 1000ML

   pH: 7.4±0.2

2. ***Bacillus cereus* Agar**

   Peptone: 10.0g

   Beef Extract: 1.0g

   D-Mannitol: 10.0g

   Sodium chloride (Nacl):5.0g

   Agar: 15.0g

   Distilled water: 1000ML

   pH:7.2±0.2

3. **Bile Esculin Agar**

   Peptone: 5.0g

   Beef Extract: 3.0g

   Bile Salts: 40.0g

   Ferric ammonium citrate: 0.5g

   Esculin: 1.0g

   Agar: 15.0g

   Distilled water: 1000ML

   pH: 6.6±0.2

4. **Mueller Hinton Agar**

   Beef Extract: 2.0g

Casein hydrolysate (acid hydrolysate of casein):17.5g

Starch: 1.5g

Agar: 17.0g

Distilled water: 1000ML

pH: 7.3±0.1